

PVN-mPFC OT projections modulates pup-directed pup care or attacking in virgin mandarin voles

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Abstract

In many species, adult animals may exhibit caregiving or aggression towards conspecific offspring. The neural mechanisms underlying the infanticide and pup care remain poorly understood. Here, using monogamous virgin mandarin voles (*Microtus mandarinus*) that may exhibit pup care or infanticide, we found that more oxytocin (OT) neurons in the paraventricular nucleus (PVN) were activated during pup caring than infanticide. Optogenetic activation of OT neurons in the PVN facilitated pup-caring in male and female mandarin voles. In infanticide voles, optogenetic activation of PVN OT cells prolonged latency to approach and attack pups, whereas inhibition of these OT neurons facilitated approach and infanticide. In addition, OT release in the medial prefrontal cortex (mPFC) in pup-care voles increased upon approaching and retrieving pups, and decreased in infanticide voles upon attacking pups. Optogenetic activation of PVN OT neuron projections to the mPFC shortened the latency to approach and retrieve pups and facilitated the initiation of pup care, whereas inhibition of these projections had little effect. For pup-care females, neither activation nor inhibition of the fibers affected their behavior towards pups. In infanticide male and female voles, optogenetic activation of PVN-mPFC OT projection fibers prolonged the latency to approach and attack pups and suppressed the initiation of infanticide, whereas inhibition of these projections promoted approach and infanticide. Finally, we found that intraperitoneal injection of OT promoted pup care and inhibited infanticide behavior. It is suggested that the OT system, especially PVN OT neurons projecting to mPFC, modulates pup-directed behaviors and OT can be used to treat abnormal behavioral responses associated with some psychological diseases such as depression and psychosis.

eLife assessment

This **important** work provides insights into the neural mechanisms regulating specific parental behaviors. By identifying a key role for oxytocin synthesizing cells in the paraventricular nucleus of the hypothalamus and their projections to the medial prefrontal cortex in promoting pup care and inhibiting infanticide, this study advances our understanding of the neurobiological basis of these contrasting behaviors in male and female mandarin voles. The evidence supporting the authors' conclusions is **solid**, and this work should be of interest to researchers studying neuropeptide control of social behaviors in the brain.

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Introduction

Both paternal and maternal care are critical to the survival as well as the physical and mental well-being of the offspring (He et al., 2019). Although we know a great deal about the neural mechanisms underlying maternal care, the neural substrates of paternal behavior remain elusive because of lacking ideal animal model of paternal care. Only in some monogamous rodents (e.g. prairie voles), canids and primates, do males assist and spend a great deal of energy caring for pups (Malcolm, 2015; Mendoza & Mason, 1986; Rosenfeld, Johnson, Ellersieck, & Roberts, 2013). However, some male rodents without reproductive experience also show paternal care toward alien pups, while some others ignore or even attack alien pups (Dai et al., 2022). These different pup-directed behavioral responses are based on their physiological and environmental states, and the killing of young of the conspecifics by sexually inexperienced mammals is a widespread phenomenon among different animal taxa (Lukas & Huchard, 2014). Infanticide is thought to benefit the infanticide by promoting their own reproduction (Hrdy, 1974). In laboratory mice, male mice without pairing experience typically display infanticide (Svare & Mann, 1981), but males are able to shift from infanticide to pup care when they have the opportunity to encounter their own offspring (Elwood, 1977). Compared to our extensive understanding of the maternal circuit, little is known about the neural substrates underlying female infanticide. The virgin mandarin voles (*Microtus mandarinus*) naturally exhibit biparental care and infanticide in both female and male that provide ideal animal model to reveal mechanism underlying paternal care in male and infanticide in females.

OT is well-known as a key hormone for initiating and maintaining maternal care (Yoshihara, Numan, & Kuroda, 2018), which is primarily synthesized in the PVN and supraoptic nucleus (SON), among which the PVN plays an important role in initiating maternal care in rats (Munetomo, Ishii, Miyamoto, Sakuma, & Kondo, 2016). There is evidence to suggest that OT not only regulates maternal motivation, but also mediates paternal behavior (Bales, Kim, Lewis-Reese, & Sue Carter, 2004). When mouse fathers were exposed to their pups, OT neurons in the PVN were specifically activated and they also showed more aggression towards intruders to protect the own pups (Shabalova et al., 2020). Compared with rodents, similar neuropeptides and hormones are involved in paternal behavior in non-human primates (Woller et al., 2012). Similar to other mammals, paternal care only exists in a few primate species (Dulac, O'Connell, & Wu, 2014; Woller et al., 2012). A study on marmoset monkeys showed that fathers have higher levels of OT secretion in the hypothalamus than non-fathers (Woller et al., 2012), and intraventricular infusion of OT reduces the tendency of marmoset fathers to refuse to transfer food to their young offspring (Saito & Nakamura, 2011). However, where and how pathways of OT neurons regulate pup care and infanticide behavior remain largely unknown.

The mPFC is involved in attention switching, decision-making, behavioral flexibility and planning, making it potentially crucial for rapidly expressing pup care or infanticide behavior. A study on virgin male and female mice found that the mPFC lesion (targeting the prelimbic cortex) significantly affected number of females and males showing pup cares and infanticide. 50% females in lesioned group exhibited maternal while 100% of sham operated groups show maternal care, whereas the 100% lesioned males exhibited infanticide, 83% of control males showed infanticide (Alsina-Llanes & Olazábal, 2021 [DOI](#)). It has been reported that the mPFC is highly activated in human mothers when they hear cry from their babies (Lorberbaum et al., 2002 [DOI](#)). The mPFC is also activated when rat mother contacts their pups first time (Fleming & Korsmit, 1996 [DOI](#)), while the damage of the mPFC disrupts pup retrieving and grooming behavior in rats (Afonso, Sison, Lovic, & Fleming, 2007 [DOI](#)). In rodents and humans, the mPFC was activated during the process of caring offspring (Hernández-González, Navarro-Meza, Prieto-Beracochea, & Guevara, 2005 [DOI](#); Seifritz et al., 2003 [DOI](#)). A study in rat mothers indicated that inactivation or inhibition of neurons in the mPFC largely reduced pup retrieval and grouping (Febo, Felix-Ortiz, & Johnson, 2010 [DOI](#)). In a subsequent study on firing patterns in the mPFC of rat mother suggested that sensory-motor processing carried out in the mPFC may affect decision making of maternal care to their pups (Febo, 2012 [DOI](#)). Examining different regions of the mPFC (anterior cingulate (Cg1), prelimbic (PrL), infralimbic (IL)) of new mother identified a role for the IL cortex in biased preference decision-making in favour of the offspring (Pereira & Morrell, 2020 [DOI](#)). A study on rats suggests that the IL and Cg1 subregion in mPFC are the motivating circuits for pup-specific biases in the early postpartum period (Pereira & Morrell, 2011 [DOI](#)), while the PrL subregion, are recruited and contribute to the expression of maternal behaviors in the late postpartum period (Pereira & Morrell, 2011 [DOI](#)). In addition, a large number of neurons in the mPFC express oxytocin receptors (OTRs) (Smeltzer, Curtis, Aragona, & Wang, 2006 [DOI](#)). OT in the circulation of mice can act on the medial prefrontal cortex (mPFC) to increase social interaction and maternal behavior (Munesue et al., 2021 [DOI](#)). Although there are some studies on the role of the mPFC in pup care, the involvement of mPFC OT projections in pup cares and infanticide requires further research. Thus, we hypothesized that PVN OT neurons projections to mPFC may causally control paternal or infanticide behaviors.

Based on the potential antagonistic effects between pup care and infanticide behavior in neural mechanisms (Dulac et al., 2014 [DOI](#); Kohl, Autry, & Dulac, 2017 [DOI](#); Mei, Yan, Yin, Sullivan, & Lin, 2023 [DOI](#)), as well as the potential role of PVN-to-mPFC OT projections in pup care and infanticide. Here, a combination of methods, including immunohistochemistry, optogenetics, fiber photometry and intraperitoneal injection of OT were used to reveal neural mechanisms underlying paternal care and infanticide. We found that PVN OT neurons regulated the expression of pup care and infanticide, and further identified the involvement of PVN-to-mPFC OT projections in paternal care and infanticide. Collectively, these findings establish a regulatory role for PVN-to-mPFC OT neurons in the expression of pup-directed behaviors and suggest potential targets for the future development of intervention strategies against psychiatric disorders associated with infanticide such as depression and psychosis.

Results

Pup care behavior activate more OT⁺ cells than infanticide in PVN

In order to observe the activated OT neurons in virgin voles during pup care and infanticide behaviors, we co-stained OT and c-Fos on brain slices from voles exhibiting different behaviors using immunofluorescence method (Fig. 1a [DOI](#)). Histological analysis showed no difference in the number of OT or c-Fos positive cells between the pup care and infanticide groups of female (Fig. 1b,c [DOI](#), Fig. 1-source data 1) and male (Fig. 1e,f [DOI](#), Fig. 1-source data 1) voles. Approximately 11% (male) and 14% (female) of OT cells expressed c-Fos during pup caring, whereas only about 3% (males) and 7% (females) of OT neurons were labeled by c-Fos during infanticide (female: $t(6) =$

5.173, $P < 0.01$, $d = 3.658$, **Fig. 1d**, Fig. 1-source data 1; male: $t(6) = -2.607$, $P < 0.05$, $d = 1.907$, **Fig. 1g**, Fig. 1-source data 1). In male and female voles, more OT neurons were activated during pup-caring than infanticide (**Fig. 1d, g**). In addition, females displaying pup care and infanticide showed higher merge rates of OT and c-Fos than males displaying the same behaviors ($F(1,12) = 5.002$, $P = 0.045$, $\eta^2 = 0.294$, **Fig. 1d, g**, Fig. 1-source data 1).

Effects of optogenetic activation of PVN

OT neurons on pup-directed behaviors

To reveal causal role of PVN OT neuron in regulation of pup-care and infanticide behaviors, effects of optogenetic activation of PVN OT neurons on pup-directed behaviors were investigated (**Fig. 2a,b,c**). Over 89% of CHR2 expression overlapped with OT neurons indicating high specificity of CHR2 virus (**Fig. 2d**, Figure 2-source data 1). 473 nm light stimulation increased c-Fos expression in CHR2 virus infected brain region that validated the effect of optogenetic activation (**Supplementary Data Fig. 1a-c**). We found that optogenetic activation of PVN OT cells significantly reduced latency to approach (CHR2: off vs on $F(1, 7) = 11.374$, $P < 0.05$, OFF/ON: $\eta^2 = 0.592$, **Fig. 2o**, Figure 2-source data 1) and retrieve pups (CHR2: off vs on $F(1, 4) = 14.755$, $P < 0.05$, OFF/ON: $\eta^2 = 0.156$, **Fig. 2p**, Figure 2-source data 1) and prolonged crouching time (CHR2: off vs on $F(1, 7) = 60.585$, $P < 0.001$, OFF/ON: $\eta^2 = 0.419$, **Fig. 2s**, Figure 2-source data 1) in pup-care males, but had no effect on females (**Fig. 2j-n**, Figure 2-source data 1), nor in control virus group. Optogenetic activation of these neurons significantly reduced the latency to approach and attack pups in male (approach: CHR2: off vs on $F(1, 5) = 185.509$, $P < 0.0001$, OFF/ON: $\eta^2 = 0.552$, **Fig. 2h**, Figure 2-source data 1; infanticide: CHR2: off vs on $F(1, 5) = 59.877$, $P < 0.01$, OFF/ON: $\eta^2 = 0.526$, **Fig. 2i**, Figure 2-source data 1) and female voles (approach: CHR2: off vs on $F(1, 6) = 64.810$, $P < 0.001$, OFF/ON: $\eta^2 = 0.915$, **Fig. 2f**, Figure source data 1; infanticide: CHR2: off vs on $F(1, 6) = 75.729$, $P < 0.001$, OFF/ON: $\eta^2 = 0.940$, **Fig. 2g**, Figure 2-source data 1) displaying infanticide behaviors whereas had no effect on control virus group (**Fig. 2f-I**, Figure 2-source data 1). Further, we conducted a two-way rmANOVA on the CHR2 data set for both sexes and found that pup-care females exhibited shorter latencies to approach (OFF: gender simple effect $F(1,13) = 5.735$, $P = 0.032$, $\eta^2 = 0.306$, **Fig. 2j, o**) and retrieve pups (OFF: gender simple effect $F(1,10) = 13.040$, $P = 0.005$, $\eta^2 = 0.566$, **Fig. 2k, p**) than males (Figure 2-source data 1). These results suggest that activation of PVN OT neurons facilitate pup care behavior and significantly inhibit infanticide behavior.

Effects of optogenetic inhibition of PVN

OT neurons on pup-directed behaviors

To further verify the roles of PVN OT neurons on pup-induced behavior, we optogenetically inhibited OT cells by eNpHR virus and tested pup-directed behaviors (**Fig. 3a,b,c**). More than 90% of neurons expressing eNpHR overlapped with OT positive neurons indicating high specificity of eNpHR virus infection (**Fig. 3d**, Figure 3-source data 1). 589 nm light stimulation to eNpHR virus infected brain regions reduced c-Fos expression verifying the effectiveness of optogenetic inhibition via eNpHR virus (**Supplementary Data Fig. 1d-f**). Inhibition of PVN OT neurons showed no significant effect on pup care behavior in male and female voles that spontaneously exhibited pup caregiving behaviors (**Fig. 3j-s**, Figure 3-source data 1). For both male and female voles in the infanticide group, optogenetic inhibition significantly shortened the latency to approach (female: $F(1, 5) = 1331.434$, $P < 0.0001$, OFF/ON: $\eta^2 = 0.980$, **Fig. 3f**, Figure 3-source data 1; male: $F(1, 5) = 10.472$, $P < 0.05$, OFF/ON: $\eta^2 = 0.690$, **Fig. 3h**, Figure 3-source data 1) and attack pups (female: $F(1, 5) = 291.606$, $P < 0.0001$, OFF/ON: $\eta^2 = 0.991$, **Fig. 3g**, Figure source data 1; male: $F(1, 5) = 46.901$, $P < 0.01$, OFF/ON: $\eta^2 = 0.837$, **Fig. 3i**, Figure 3-source data 1). In addition, we performed a two-way rmANOVA on the eNpHR group data for both sexes and found that pup-care females exhibited shorter latency to approach (gender main effect $F(1,10) = 62.131$, $P < 0.0001$,

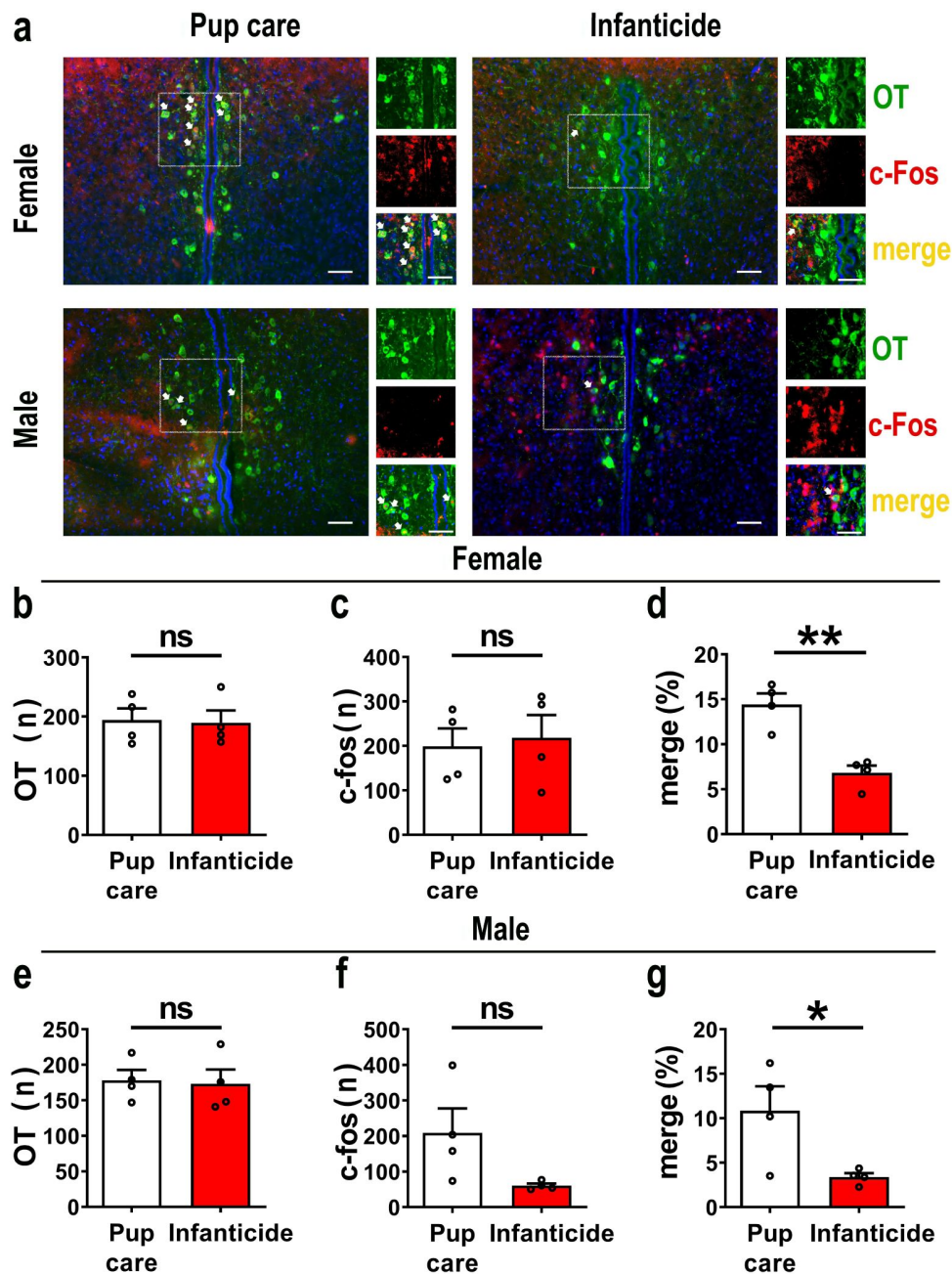


Figure 1

Activated OT neurons in the PVN of mandarin voles during pup care (n = 4) and infanticide (n = 4).

a. Representative histological images of OT (green) and c-Fos (red) positive cells in PVN: Blue, DAPI; Yellow, merged cells. Enlarged views of the boxed area were shown to the right of each image, white arrows indicated the overlap of OT and c-fos positive cells. Objective: 20x. Scale bars, 50 μ m. b. Number of OT positive cells in pup care and infanticide female voles. c. Number of c-Fos positive cells in pup care and infanticide female voles. d. Percentage of c-Fos-expressing cells in OT cells of PVN from pup care and infanticide female voles. ****** $P < 0.01$. Independent sample t-tests. e. Number of OT cells in pup care and infanticide male voles. f. Number of c-Fos cells in pup care and infanticide male voles. g. Percentage of c-Fos-expressing cells in OT cells of PVN from pup care and infanticide male voles. ***** $P < 0.05$. Independent sample t-tests. Error bars indicate SEM.

Figure 1-source data 1.

Statistical results of the number of OT-positive cells, the number of c-Fos-positive cells, and the percentage of OT and c-Fos merged neurons in the total PVN OT neurons in female and male pup-care and infanticide voles.

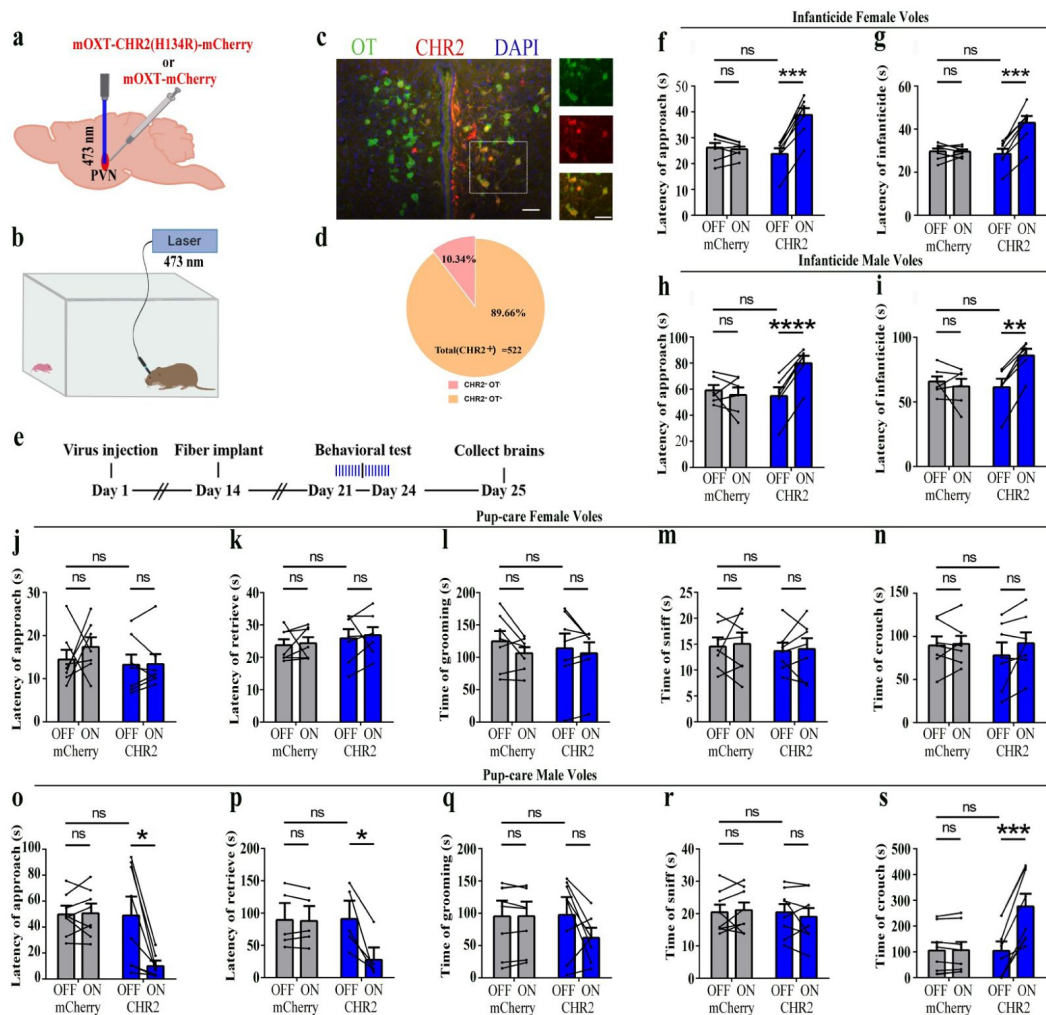


Fig. 2

Effects of optogenetic activation of PVN OT neurons on pup-directed behaviors

a. Schematic of virus injection and optical fiber implantation. b. Schematic diagram of the behavioral test. c. Representative histologic images of CHR2 (red) expression and OT staining (green), enlarged view of the boxed area is on the right side. Blue, DAPI. Objective: 20x. Scale bars, 50 μ m. d. Statistics on the specificity of CHR2 expression in 3 voles, more than 89% of CHR2 positive neurons overlapped with OT positive neurons. e. Time line of the experiment. f-i, Approach (f, h) and infanticide (g, i) latency of infanticide 7 female and 6 male voles in mCherry (control virus group) and CHR2 groups. $^{**}P < 0.01$ vs. CHR2 OFF; $^{***}P < 0.001$ vs. CHR2 OFF; $^{****}P < 0.0001$ vs. CHR2 OFF. Two-way rmANOVA (factors: treatment $\times$ stimulus). j-n, Latency to approach (j), latency to retrieve (k), grooming time (l), sniffing time (m) and crouching time (n) of 7 pup-care female voles in female control virus and CHR2 groups. o-s, Latency to approach (o), latency to retrieve (p), grooming time (q), sniffing time (r) and crouching time (s) of 8 pup-care male voles in male control virus and CHR2 groups. $^{*}P < 0.05$ vs. CHR2 OFF, $^{***}P < 0.001$ vs. CHR2 OFF. Two-way rmANOVA (factors: treatment $\times$ stimulus). Error bars indicate SEM.

Figure 2-source data 1.

Statistical results of the number of cells expressing only CHR2 and co-expressing CHR2 and OT in the PVN of 3 voles with injection of optogenetic virus, the latency to approach and attack pups in infanticide voles, and the latency to approach, retrieve, duration of grooming, sniffing, and crouching in pup-care voles.

$\eta^2 = 0.861$, **Fig. 3j, o** [↗](#)) and retrieve (gender main effect $F(1,10) = 137.393$, $P < 0.0001$, $\eta^2 = 0.932$, **Fig. 3k, p**) than males (Figure 3-source data 1). These results suggest that inhibition of OT neurons in the PVN significantly facilitates infanticide behavior.

Changes in OT release upon pup-directed behaviors

The results of the optogenetic manipulation demonstrated that PVN OT neurons regulated pup-induced behavior. We next detected the OT release in the mPFC during pup-induced behavior by OT1.0 sensor (**Fig. 4a-c** [↗](#)). Pup-caring female and male voles showed a significant increase (female: $F(1.958, 13.708) = 45.042$, $P < 0.001$, $\eta^2 = 0.865$; male: $F(5, 35) = 24.057$, $P < 0.01$, $\eta^2 = 0.775$) in the signal for OT1.0 sensors upon approaching (female: $P < 0.01$; male: $P < 0.05$) and retrieving (female: $P < 0.01$; male: $P < 0.05$), whereas there was no significant difference in the signal at the onset of other behaviors compared with the signal before the introduction of the pups (**Fig. 4f-m** [↗](#), Figure 4-source data 1). In addition, we compared the signals of OT1.0 sensors in the pup-caring voles upon the first, second and third approaches to the pups (female: $F(2, 14) = 10.917$, $P < 0.01$, $\eta^2 = 0.609$; male: $F(2, 14) = 13.351$, $P < 0.01$, $\eta^2 = 0.656$), OT release peaked at the first approach and tended to decrease thereafter (**Fig. 4d,e** [↗](#), Figure 4-source data 1). In infanticide female and male voles, OT release decreased upon attacking in infanticide males ($F(1.117, 7.822) = 85.803$, $P < 0.001$, $\eta^2 = 0.838$) and females ($F(1.068, 7.479) = 36.336$, $P < 0.001$, $\eta^2 = 0.925$) (**Fig. 4n-r** [↗](#), Figure 4-source data 1). In addition, no significant changes in signals were detected from the individuals with control AAV2/9-hSyn-OTmut during pup-directed behaviors (**Supplementary Data Fig. 2** [↗](#)). In addition, no changes in OT release were detected while subjects were exposed to object with similar size and shape as the pup (**Supplementary Data Fig. 3** [↗](#)). Besides, we found that pup-care females showed higher AUC per second than males during approaching (gender simple effect $F(1,14) = 27.740$, $P = 0.000119$, $\eta^2 = 0.665$, **Fig. 4l, m** [↗](#), Figure 4-source data 1) and retrieving (gender simple effect $F(1,14) = 11.695$, $P = 0.004$, $\eta^2 = 0.455$, **Fig. 4l, m** [↗](#), Figure 4-source data 1) the pups. These results indicate that mPFC OT release significantly increased upon approaching and retrieving in pup-care voles, but decreased upon attacking pups in infanticide voles.

Effects of optogenetic activation of PVN OT neurons fibers in the mPFC on pup-directed behaviors

Previous experiment found that OT release in the mPFC changed upon pup-directed behavior. To manipulate the neural circuit, we first verified oxytocin projections from PVN to the mPFC. We injected retrogradely labeled virus in the mPFC and observed the overlap of virus with OT in the PVN (**Fig. 5** [↗](#)), and we also counted the PVN OT neurons projecting to mPFC and found that approximately 45.16% and 40.79% of cells projecting from PVN to the mPFC were OT-positive, and approximately 18.48% and 18.89% of OT cells in the PVN projected to the mPFC in females and males, respectively (**Supplementary Data Fig. 4** [↗](#)). Then, we tested whether optogenetic activation of the PVN OT neuron projection fibers in the mPFC affects pup-induced behaviors (**Fig. 6a-d** [↗](#)). Similar to previous results, in the pup caring group, activation of the fibers facilitated approaching ($F(1,5) = 23.915$, $P < 0.01$, OFF/ON: $\eta^2 = 0.760$) and retrieving ($F(1,5) = 39.664$, $P < 0.01$, OFF/ON: $\eta^2 = 0.907$) in male voles (**Fig. 6o-s** [↗](#), Figure 6-source data 1), but had no effect on females (**Fig. 6j-n** [↗](#), Figure 6-source data 1). In male and female infanticide group voles, optogenetic activation of the PVN OT neuron projection fibers prolonged latency to approach (female: $F(1,5) = 37.094$, $P < 0.01$, OFF/ON: $\eta^2 = 0.875$; male: $F(1,5) = 74.718$, $P < 0.001$, OFF/ON: $\eta^2 = 0.889$) and attack pups (female: $F(1,5) = 38.347$, $P < 0.01$, OFF/ON: $\eta^2 = 0.877$; male: $F(1,5) = 61.589$, $P < 0.01$, OFF/ON: $\eta^2 = 0.910$) (**Fig. 6f-i** [↗](#), Figure 6-source data 1). These results suggest that activation of the PVN OT neurons to mPFC projection promoted the onset of pup care behavior in pup-care male voles and inhibited the onset of infanticide behavior in infanticide voles.

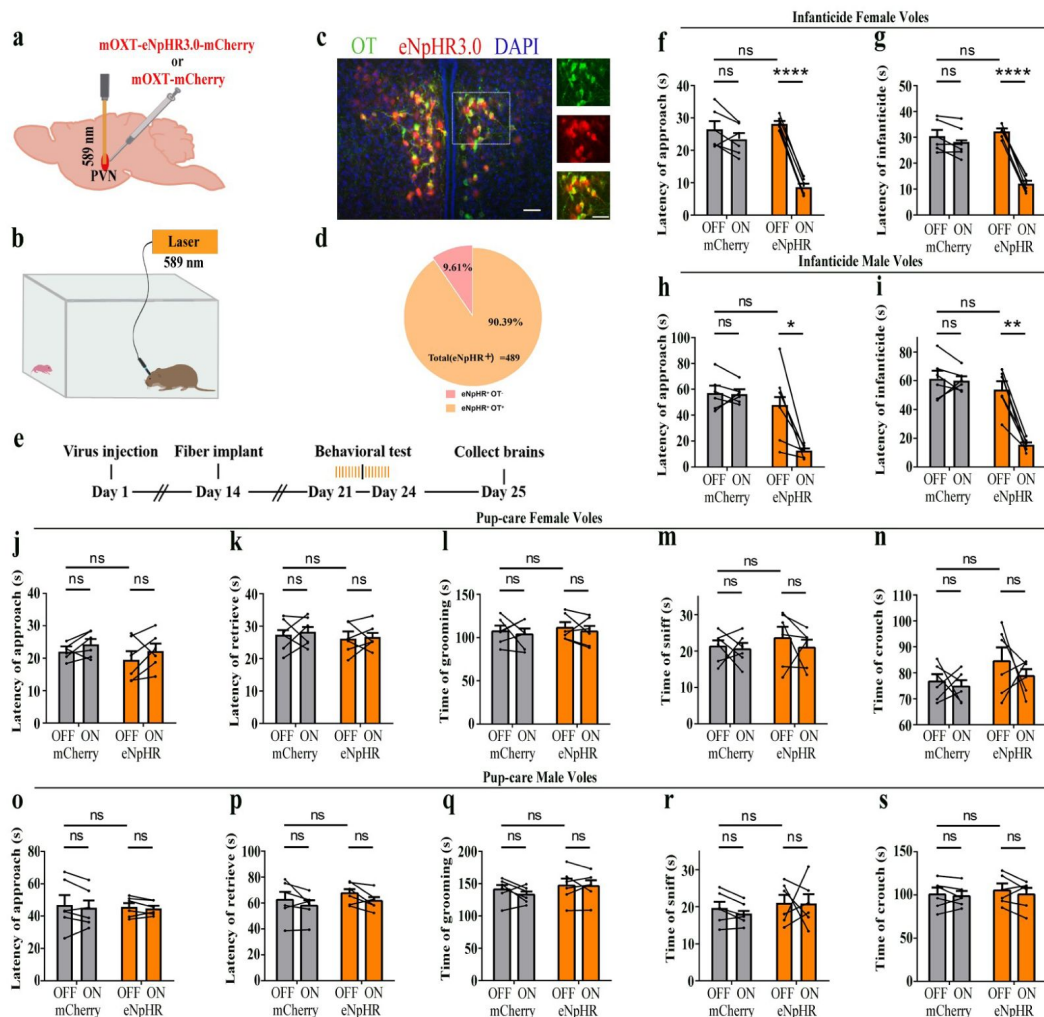


Fig. 3

Effects of optogenetic inhibition of PVN OT neurons on pup-directed behaviors

a. Schematic of virus injection and optical fiber implantation. b. Schematic diagram of the behavior test. c. Representative histological images of OT staining (green) and eNpHR (red) expression, enlarged view of the boxed area is shown on the right side. Blue, DAPI. Objective: 20x. Scale bars, 50 μ m. d. Statistics on the specificity of CHR2 expression in 3 voles, more than 90% of eNpHR expression overlapped with OT. e. Time line of the experiment. f–i, Approach (f, h) and infanticide (g, i) latency in 6 female (f, g) and 6 male (h, i) infanticide voles. * $P < 0.05$ vs. eNpHR OFF; ** $P < 0.01$ vs. eNpHR OFF; **** $P < 0.0001$ vs. eNpHR OFF. Two-way rmANOVA (factors: treatment $\times$ stimulus). j–n, Approach latency (j), retrieval latency (k), grooming time (l), sniffing time (m) and crouching time (n) in 6 control mCherry and eNpHR group of pup-care female voles. o–s, Approach latency (o), retrieval latency (p), grooming time (q), sniffing time (r) and crouching time (s) in 6 control mCherry and eNpHR group of pup-care male voles. Error bars indicate SEM.

Figure 3-source data 1.

Statistical results of the number of cells expressing only eNpHR and co-expressing eNpHR and OT in the PVN of 3 voles with injection of optogenetic virus, the latency to approach and attack pups in infanticide voles, and the latency to approach, retrieve pups, duration of grooming, sniffing, and crouching in pup-care voles.

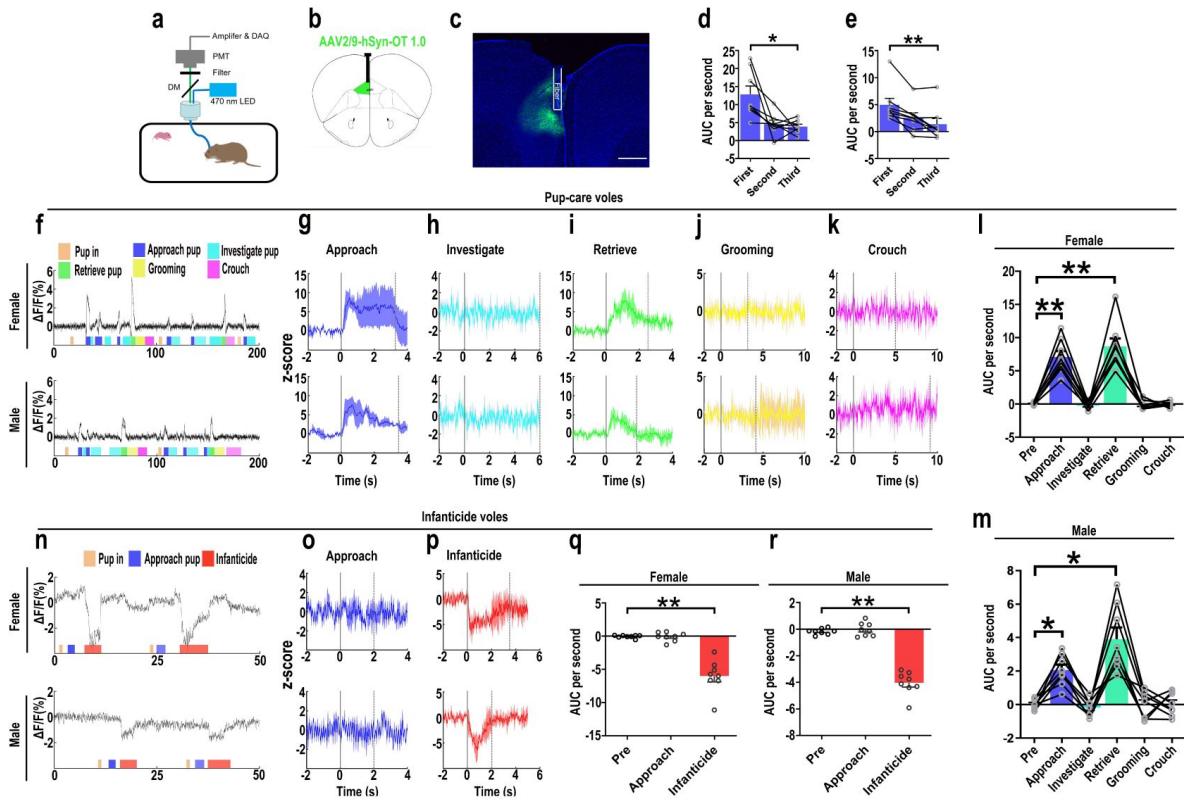


Fig. 4

OT release in the mPFC upon pup-directed behaviors

a. Recording instrument settings. b. Illustrations of viral expression and optical fibre location. c. Representative histological image of OT1.0 sensor (green) and optical fibre locations. Blue, DAPI. Objective: 4x. Scale bars, 500 μm . d, e, Area under the curve (AUC) per second for pup-care female (d) and male (e) voles approaching pups for the first, second and third time ($n = 8$). * $P < 0.05$ vs. first. ** $P < 0.01$ vs. first. One-way rmANOVA. f. Representative $\Delta F/F$ traces in pup-care female (f, top) and male (f, bottom) voles during interaction with pups. g-k, Post-event histograms (PETHs) of z-score of OT1.0 sensor for the following pup-directed behaviors: approach (g), investigate (h), retrieve (i), grooming (j) and crouch (k). l, m, The mean AUC of z-scores for pup-care female (l) and male (m) voles across various pup-directed behaviors ($n = 8$). Female: ** $P < 0.01$ vs approach. $P < 0.01$ vs retrieve. Male: * $P < 0.05$ vs approach. $P < 0.05$ vs retrieve. One-way rmANOVA. n. Representative $\Delta F/F$ traces in infanticide female (n, top) and male (n, bottom) voles during interaction with pups. o, p, PETHs of z-score of OT1.0 sensor for approach and infanticide in infanticide voles. q, r, The mean AUC of z-score of OT1.0 sensor for pre-pup exposure, approach and infanticide in infanticide female (q) and male (r) voles ($n = 8$). ** $P < 0.01$ vs infanticide. One-way rmANOVA. Error bars indicate SEM.

Figure 4-source data 1.

Area under the curve per second for pre-pup exposure, approach, and infanticide behaviors in infanticide voles and area under the curve per second for first, second, and third approaches to pups, as well as pre-pup exposure, approach, investigation, retrieval, grooming, and crouching behaviors in pup-care voles.

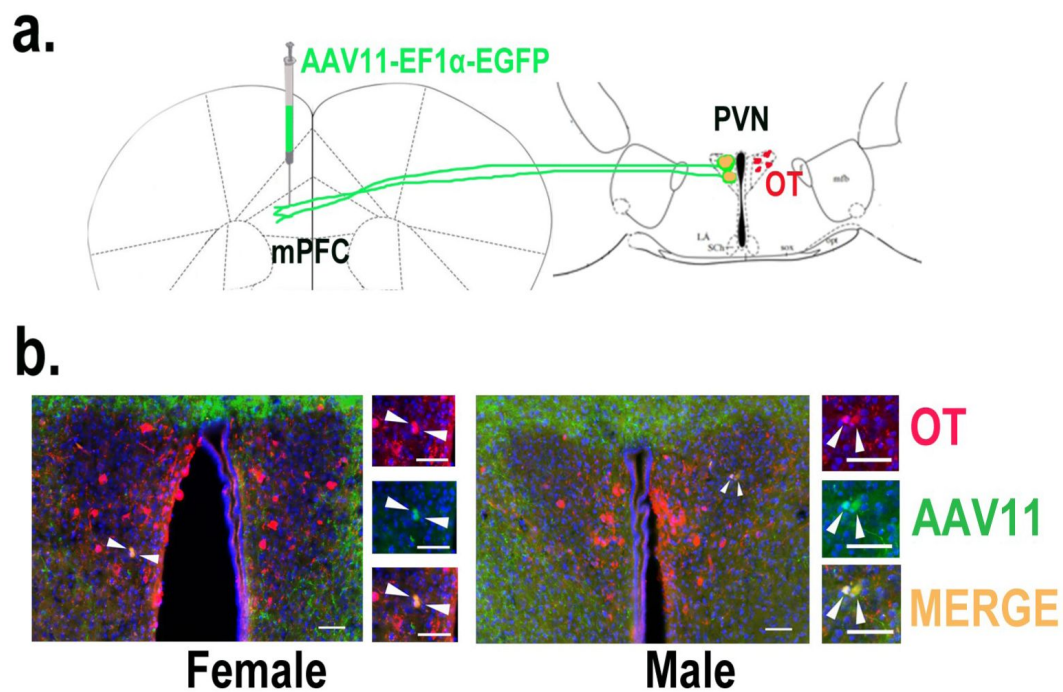


Fig. 5

Determination of PVN to mPFC oxytocin projection.

a. Schematic diagram of mPFC virus injection and OT staining. b. Histological pictures of OT (red) and AAV11 (green) co-staining in male and female. Yellow, merged. Blue, DAPI. Objective: 20x. Scale bars, 50 μ m.

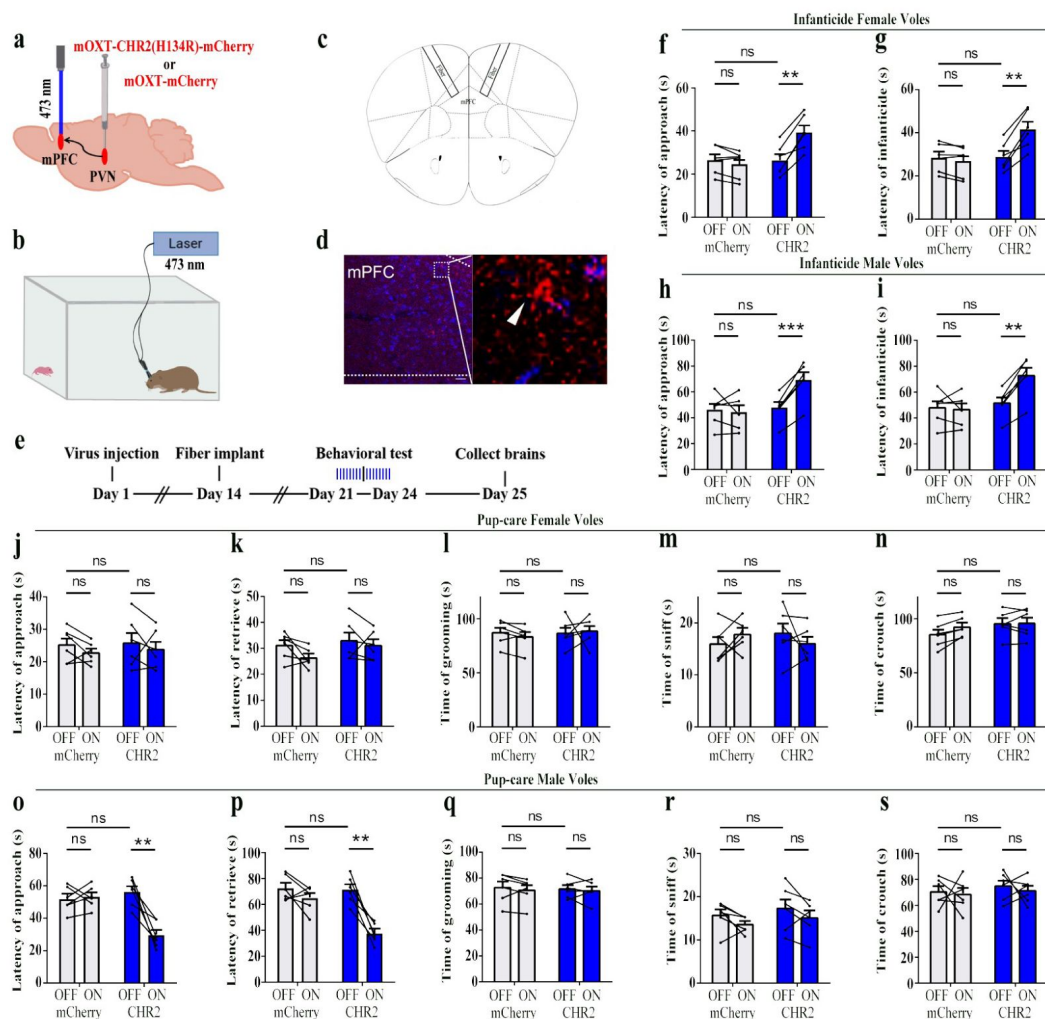


Fig. 6

Effects of optogenetic activation of the PVN OT neuron projection fibers on pup-directed behaviors

a. Schematic of virus injection and optical fiber implantation. b. Schematic diagram of the behavior test. c. Illustration of optical fiber implantation in the target brain region. d. Representative histological pictures of the fiber position and projection fibers. Blue, DAPI. Objective: 20x. Scale bars, 50 μ m. e. Time line of the experiment. f-i, Changes in approach latency (f, h) and infanticide latency (g, i) of female (f, g) and male (h, i) infanticide voles in CHR2 and control mCherry group before and after delivery of light ($n = 6$). $**P < 0.01$ vs. CHR2 OFF; $***P < 0.001$ vs. CHR2 OFF. Two-way ANOVA (factors: treatment $\times$ stimulus). j-n, Approach latency (j), retrieval latency (k), grooming time (l), sniffing time (m) and crouching time (n) in control mCherry and CHR2 group of pup-care female voles ($n = 6$). o-s, Approach latency (o), retrieval latency (p), grooming time (q), sniffing time (r) and crouching time (s) in control mCherry and CHR2 group of pup-care male voles ($n = 6$). $**P < 0.01$ vs. CHR2 OFF. Two-way rmANOVA (factors: treatment $\times$ stimulus). Error bars indicate SEM.

Figure 6-source data 1.

Statistical results of the latency to approach and attack pups in infanticide voles, and the latency to approach, retrieve pups, duration of grooming, sniffing, and crouching in pup-care voles.

Optogenetic inhibition of the PVN OT neuron projection fibers promoted infanticide

We then optogenetically suppressed the projection fibers from PVN OT neurons to mPFC and observed changes in pup-directed behaviors (**Fig. 7a-d**). Similar to the results of the PVN OT neurons inhibition, we found that optogenetic inhibition of the PVN OT neuron projection fibers promoted approach ($F(1,5) = 119.093$, $P < 0.001$, OFF/ON: $\eta^2 = 0.877$) and infanticide ($F(1,5) = 112.501$, $P < 0.001$, OFF/ON: $\eta^2 = 0.885$) in infanticide male (**Fig. 7h,i**, Figure 7-source data 1) and female voles (approach: $F(1,5) = 280.031$, $P < 0.0001$, OFF/ON: $\eta^2 = 0.853$; infanticide: $F(1,5) = 268.694$, $P < 0.0001$, OFF/ON: $\eta^2 = 0.838$) (**Fig. 7f,g**, Figure 7-source data 1). For pup-care male and female voles, inhibition of these fibers did not significantly affect their pup care behaviors (**Fig. 7j-s**, Figure 7-source data 1). To validate the effectivity of fiber optogenetic inhibition, we combined optogenetic inhibition with OT1.0 sensors and recorded a decrease in OT release upon inhibition of fibers (**Supplementary Data Fig. 5**). These results suggest that optogenetic inhibition of the PVN OT neuron projection fibers promotes the onset of infanticide behavior in infanticide voles.

Intraperitoneal injection of OT

For pre-clinic purpose, we tested the effect of peripheral administration of OT on pup-directed behavior (**Fig. 8a**). Delivery of OT promoted approach ($t(4) = 3.737$, $P < 0.05$, $d = 1.335$) and retrieval ($t(4) = 4.190$, $P < 0.05$, $d = 2.04$) in pup-care male voles (**Fig. 8k, m**), while it had no significant effect on the pup-directed behaviors in females (**Fig. 8f-j**). For infanticide voles, there was not significant prolongation of approach latency (**Fig. 8d**), but a significant extension of infanticide latency in males after delivery of OT ($t(6) = -2.988$, $P < 0.05$, $d = 1.345$, **Fig. 8e**). In infanticide female voles, both the latency to approach ($Z = -2.380$, $P < 0.05$, $d = 5.891$, **Fig. 8b**) and attack pups ($t(7) = -3.626$, $P < 0.01$, $d = 1.063$, **Fig. 8c**) were significantly prolonged by the delivery of OT. In addition, we integrated data from our pre-test of effects of OT on numbers of infanticide voles and we found that intraperitoneal injection of OT significantly reduced the number of infanticide female ($\chi^2 = 4.740$, $P < 0.05$, odds ratio: OR = 0.303, **Fig. 8p**) and male voles ($\chi^2 = 3.039$, $P = 0.081$, OR = 0.292, **Fig. 8q**). These results indicate that peripheral delivery of OT can promote the onset of pup care behavior in pup-care male voles and can significantly suppress the infanticide in both sexes. This provides a basis for the application of OT in clinic and wild life management.

Discussion

In this study, we used monogamous, highly social mandarin voles to explore the role of PVN OT neurons and PVN^{OT}-mPFC projections in the modulation of pup care and infanticide behaviors. More OT neurons in the PVN were activated during pup care than infanticide behaviors. Optogenetic activation of the OT neurons in the PVN or OT neuron fibers in the mPFC promoted pup care and inhibited infanticide behavior, whereas inhibition of these neurons and their fibers in the mPFC promoted infanticide. In addition, intraperitoneal administration of OT promoted approach and retrieval of pups in pup care male voles, and inhibited infanticide in both male and female voles. The present study revealed that the PVN to mPFC OT neural projections regulate pup care and infanticide behavior in virgin mandarin voles.

Firstly, we found that more OT neurons in the PVN were activated during pup care than infanticide behaviors, which is consistent with the well-established prosocial role of OT and its ability to promote pup care behavior (Bosch & Young, 2018). In a previous study on virgin male prairie voles, OT and Fos colabeled neurons in PVN increased after exposure to conspecific pups

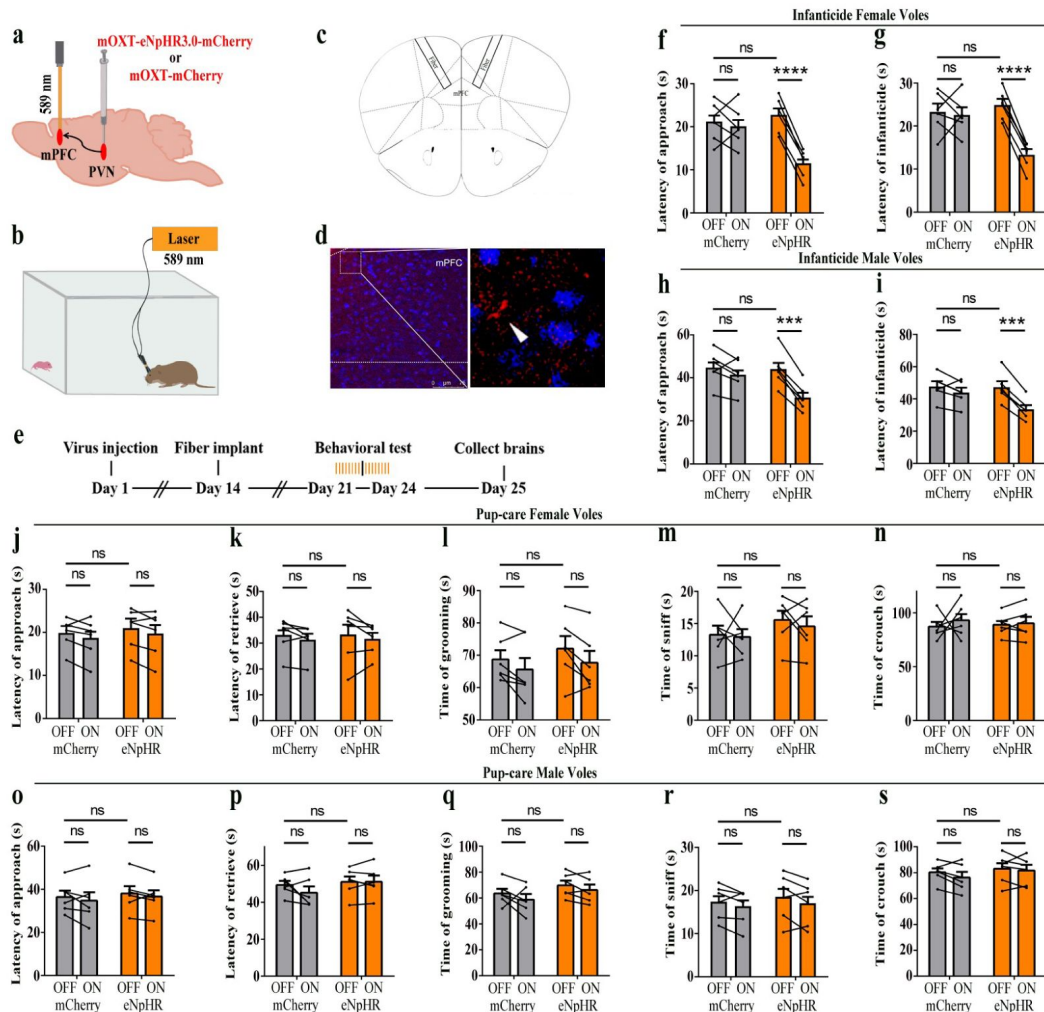


Fig. 7

Optogenetic inhibition of the PVN OT neuron projection fibers promoted the onset of infanticide

a. Illustration of virus injection and optical fiber implantation. b. Schematic of the behavior test. c. Diagram of optical fiber implantation in the target brain region. d. Representative histological pictures of the fiber location and projection fibers. Blue, DAPI. Objective: 20x. Scale bars, 75 μ m. e. Time line of the experiment. f-i, Changes in approach (f, h) and infanticide (g, i) latency of female and male infanticide voles in eNpHR (n = 6) and control mCherry groups (n = 6) before and after light delivery. $***P < 0.001$ vs. eNpHR OFF. $****P < 0.0001$ vs. eNpHR OFF. Two-way rmANOVA (factors: treatment $\times$ stimulus). j-n, Approach latency (j), retrieval latency (k), grooming time (l), sniffing time (m) and crouching time (n) in control mCherry (n = 6) and eNpHR (n = 6) group of pup-care female voles. o-s, Approach latency (o), retrieval latency (p), grooming time (q), sniffing time (r) and crouching time (s) in control mCherry (n = 6) and eNpHR group (n = 6) of pup-care male voles. Error bars indicate SEM.

Figure 7-source data 1.

Statistical results of the latency to approach and attack pups in infanticide voles, and the latency to approach and retrieve pups, duration of grooming, sniffing, and crouching in pup-care voles.

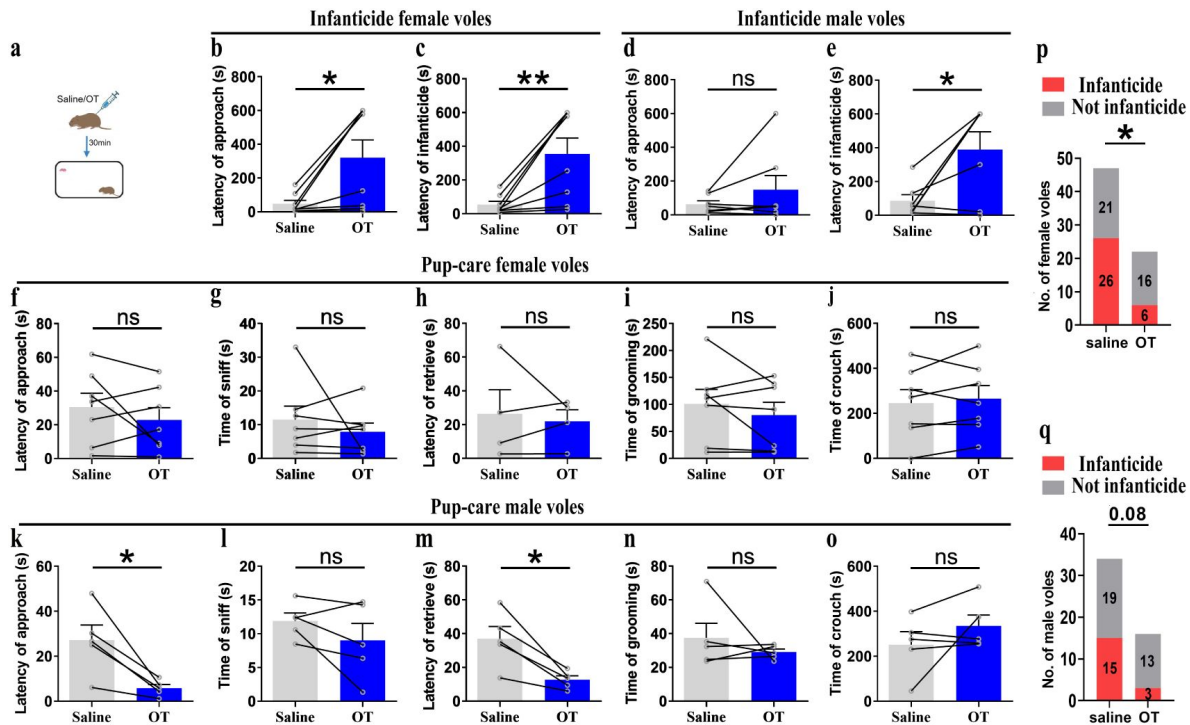


Fig. 8

Pup-directed behaviors before and after intraperitoneal delivery of OT

a. Diagram of intraperitoneal OT delivery. b,c, Approach (b) and infanticide latency (c) of infanticide female voles ($n = 8$). $*P < 0.05$. Paired-samples t-test. $**P < 0.01$. Wilcoxon signed ranks test. d,e, Approach (d) and infanticide latency (e) of infanticide male voles ($n = 7$). $*P < 0.05$. Paired-samples t-test. f-j, Approach latency (f, $n = 7$), sniffing time (g, $n = 7$), latency to retrieve (h, $n = 4$), grooming time (i, $n = 7$) and crouching time (j, $n = 7$) before and after delivery of OT in pup-care female voles. k-o, Approach latency (k), sniffing time (l), latency to retrieve (m), grooming time (n) and crouching time (o) before and after delivery of OT in pup-care male voles ($n = 5$). $*P < 0.05$. Paired-samples t-test. p,q Changes in infanticide rates in female (p) and male (q) voles after administration of saline and OT. $*P < 0.05$. Pearson chi-square test. Error bars indicate SEM. Figure 8-source data 1.

Statistical results of the latency to approach and attack pups in infanticide voles, and the latency to approach and retrieve pups, duration of grooming, sniffing, and crouching in pup-care voles. Statistical results on the number of infanticide or non infanticide voles induced by injection of saline and OT.

and experiencing paternal care (Kenkel et al., 2012 [↗](#)). In another study of prairie voles, OT and c-Fos colabeled neurons in the PVN significantly increased after becoming parents, which may be due to a shift from virgin to parents (Kelly, Hiura, Saunders, & Ophir, 2017 [↗](#)). Meanwhile, we found that activation of OT neurons in the PVN facilitated pup care behaviors such as approach, retrieval and crouching in pup-care male voles; whereas inhibition of OT had no effect on paternal behavior; activation and inhibition of OT neurons in the PVN had no significant effect on pup care behaviors in pup-care females; and activation of OT neurons in the PVN inhibited pup killing in infanticide voles, whereas the corresponding inhibition of OT neurons in the PVN facilitated infanticide. This finding is consistent with previous report that silencing OT neurons delayed the retrieval behavior in virgin mice (Carcea et al., 2021 [↗](#)). Further study found that simply observing dams retrieve pups through a transparent barrier could increase retrieval behavior and PVN OT neuron activity of virgin females (Carcea et al., 2021 [↗](#)). If OTR knockout mice are used, no pup retrieval occurs after observation, and these results suggest that activation of PVN OT neurons in virgin mice induced by visual signals promoted pup care behaviors (Carcea et al., 2021 [↗](#)). In addition, this study further demonstrated that OT in the PVN facilitated the retrieval behavior by modulating the plasticity of the left auditory cortex and amplifying the response of mice to the pup's call (Carcea et al., 2021 [↗](#)). In our another study, we found that the OT neurons in the PVN projecting to the VTA as well as to the Nac brain region regulate pup-directed behaviors, which may also be accompanied by dopamine release (He et al., 2021 [↗](#)). This studies also support finding from optogenetic activation of OT neurons in the PVN in the present study. The results of the present study are also supported by a recent study that chemogenetic activation OT neurons in the PVN increase pup care and reduce infanticide (Inada et al., 2022 [↗](#)).

However, manipulation of OT neurons in the PVN produced no significant effect on pup caring in pup-care females. This may be due to the fact that female voles have inherently higher OT neural activity (Häussler, Jirikowski, & Caldwell, 1990 [↗](#)), and female mice have more OT neurons and OT axon projections than males (Häussler et al., 1990 [↗](#)), and that there are also significant differences in OTR expression between two sexes (Insel, Gelhard, & Shapiro, 1991 [↗](#); Uhl-Bronner, Waltisperger, Martínez-Lorenzana, Condes Lara, & Freund-Mercier, 2005 [↗](#)) that possibly shows ceiling effects of the OT system. In the present study, we found that females have more activated OT neurons (**Figure 1, d, g** [↗](#)) and released higher levels of OT into the mPFC (**Figure 4 d, e** [↗](#)) than males. This sex difference has been reported in other study that activation of OT neurons in the PVN activated noradrenergic neurons in the locus coeruleus by co-releasing OT and glutamate, increased attention to novel objects in male rats, and that this neurotransmission was greater in males than in females (Wang, Escobar, & Mendelowitz, 2021 [↗](#)). In a study on virgin female mice, pup exposure was found to activate oxytocin and oxytocin receptor expressing neurons (Okabe et al., 2017 [↗](#)). Virgin female mice repeatedly exposed to pups showed shorter retrieval latencies and greater c-Fos expression in the preoptic area (POA), concentrations of OT in the POA were also significantly increased, and the facilitation of alloparental behavior by repeated exposure to pups occurred through the organization of the OT system (Okabe et al., 2017 [↗](#)). In the present study, we also observed that optogenetic activation of OT neurons in PVN increased crouching behavior in the pup-care male voles, but did not affect grooming time possibly via increase of OT release. This result is in line with a previous study that injection of an OTR antagonist into the MPOA of male voles significantly reduced the total duration of pup care behavior and increased the latency to approach pups and initiate paternal behavior in male voles (Yuan et al., 2019 [↗](#)). This finding suggests that the results of peripherally administered OT and optogenetically activated PVN OT neurons in the present study may also have the involvement of OT-OTR interactions in MPOA. Parturition in experimental animals is accompanied by a decrease in infanticide and the emergence of pup-care behavior, and along with this process OTR expression increased not only on mammary contractile cells, but also in various regions of the brain, such as MPOA, VTA, and OB, which are all considered to be important brain regions related to the onset and maintenance of pup care behaviors (Bosch & Neumann, 2012 [↗](#); Shahrokh, Zhang, Diorio, Gratton, & Meaney, 2010 [↗](#); Yu, Kaba, Okutani, Takahashi, & Higuchi, 1996 [↗](#)). For example, in a OB study, intra-OB injection of OT antagonist significantly delayed the onset of pup care behaviors such as retrieving

pups, crouching, and nesting in female rats, whereas intra-OB injection of OT in virgin females induced 50% of females to show intact pup care behaviors (Yu et al., 1996 [\[1\]](#)). Therefore, the effects of activation of PVN OT neurons may be a result of actions on multiple brain regions involved in the expression of pup care behavior. Which brain regions that PVN OT neurons project to are involved in pups caring or infanticide needs further studies. Although we used a virus strategy to specifically activate or inhibit PVN OT neurons, other neurochemical may also be released during optogenetic manipulations because OT neurons may also release other neurochemicals. In one of our previous studies, activation of the OT neuron projections from the PVN the VTA as well as to the Nac brain also altered pup-directed behaviors, which may also be accompanied by dopamine release (He et al., 2021 [\[2\]](#)). In addition, back-propagation of action potentials during optogenetic manipulations may also causes the same behavioral effect as direct stimulation of PVN OT cells. These indirect effects on pup-directed behaviors should also be investigated further in the future study.

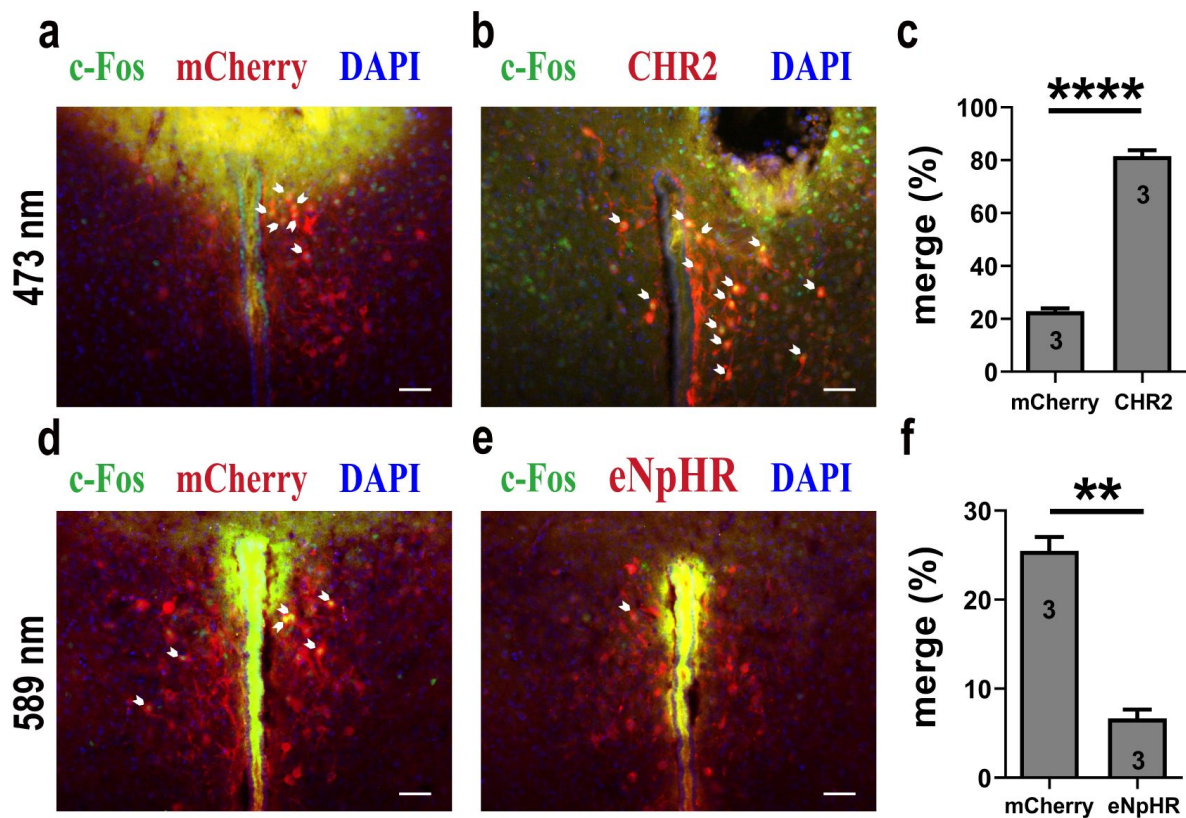
Optogenetic activation of the OT neural projection fibers from the PVN to the mPFC facilitated the onset of pup care behaviors, such as approach and retrieval, in pup-care male voles, whereas inhibition of this circuit had no effect; neither activation nor inhibition had a significant effect in pup-care females; and activation of this neural circuit inhibited infanticide in infanticide voles, whereas the corresponding inhibition of this circuit facilitated infanticide. In addition, we have demonstrated an increase in OT release in pup-care voles upon approaching and retrieving pups, and a decrease in OT release in infanticide voles when infanticide occurs, as recorded by OT sensors in the mPFC. There is some evidence indicating that the mPFC may be an important brain region for OT to exert its effects. In addition to expressing OTRs in the mPFC (W. Liu, Pappas, & Carter, 2005 [\[3\]](#); Smeltzer et al., 2006 [\[4\]](#)), the mPFC contains OT-sensitive neurons (Ninan, 2011 [\[5\]](#)) and receives projections of OT neurons from the hypothalamus (Knobloch et al., 2012 [\[6\]](#); Sofroniew, 1983 [\[7\]](#)). It has been further shown that blocking OTR in the mPFC of postnatal rats by an OTR antagonist delayed the retrieval of pups and reduced the number of pups retrieved by rats, impaired the care of pups, and decreased the latency to attack intruders, increased the number of attacks, and increased anxiety in postnatal rats but had no effect on the level of anxiety in virgin rats (Sabihi, Dong, Durosko, & Leuner, 2014 [\[8\]](#)). These results further suggest that OT in the mPFC is involved in the regulation of pup care behavior and support the results of this study in pup care voles and infanticide voles from an OTR perspective. Previous studies have shown that optical activation of the mPFC maintains aggression within an appropriate range, with activation of this brain region suppressing aggression between male mice and inhibition of the mPFC resulted in quantitative and qualitative escalation of aggression (Takahashi, Nagayasu, Nishitani, Kaneko, & Koide, 2014 [\[9\]](#)), which is similar to our findings with infanticide voles. For pup care behavior it has been reported that the mPFC brain region may be involved in the rapid initiation of pup care behavior in mice without pairing experience (Alsina-Llanes & Olazábal, 2020 [\[10\]](#)), which supported the experimental results of pup-care male voles in the present study, but the present study further suggested that OT neurons projecting to mPFC regulated pup-caring and infanticide behavior possibly via increase of OT release in the mPFC. We must also mention the function of the mPFC subregion. In the present study, virus were injected into the PrL. The PrL and IL regions of the mPFC play different roles in different social interaction contexts (Bravo-Rivera, Roman-Ortiz, Brignoni-Perez, Sotres-Bayon, & Quirk, 2014 [\[11\]](#); Moscarello & LeDoux, 2013 [\[12\]](#)). A study has shown that the PrL region of the mPFC contributes to active avoidance in situations where conflict needs to be mitigated, but also contributes to the retention of conflict responses for reward (Capuzzo & Floresco, 2020 [\[13\]](#)). This may reveal that the suppression of infanticide by PVN to mPFC OT projections is a behavioral consequence of active conflict avoidance. In a study of pain in rats, OT projections from the PVN to the PrL were found to increase the responsiveness of cell populations in the PrL, suggesting that OT may act by altering the local excitation-inhibition (E/I) balance in the PrL (Y. Liu et al., 2023 [\[14\]](#)). A study of anxiety-related behaviors in male rats suggests that the anxiolytic effects of OT in the mPFC are PL-specific and that this is achieved primarily through the

engagement of GABAergic neurons, which ultimately modulate downstream anxiety-related brain regions, including the amygdala (Sabihi, Dong, Maurer, Post, & Leuner, 2017 [DOI](#)). This may provide possible downstream pathways for further research.

Another interesting finding is that peripheral OT administration promoted pup care behavior in male voles, such as approaching and retrieving pups, but had no significant effect on pup care behaviors in female voles. This result was supported by previous study that OT administered peripherally inhibited infanticide in pregnant and reproductively inexperienced females and promoted pup caring in pregnant females (McCarthy, Bare, & vom Saal, 1986 [DOI](#)). Further research has demonstrated that OT in the central nervous system also inhibited infanticide in female house mice (McCarthy, 1990 [DOI](#)). In addition, peripheral administration of OT inhibits infanticide in male mice without pairing experience (Nakahara et al., 2020 [DOI](#)). Moreover, the experience of pairing facilitated the retrieval of pups by increasing peripheral OT levels in both male and female mice (Lopatina et al., 2011 [DOI](#)). This is similar to the findings of pup-care males in the present study, whereas the absence of similar results in females may be due to differences in OT levels between the sexes (Tamborski, Mintz, & Caldwell, 2016 [DOI](#)). Pup-care female voles in this study already showed short latency to approach and retrieve pups before peripheral administration of OT. We also found that peripheral OT administration suppressed infanticide behavior in both male and female voles. It is consistent with previous study that infanticide in female house mice with no pairing experience and pregnant females with pre-existing infanticide was effectively suppressed by subcutaneous injection of OT, while injection of OT also helped pregnant females to show care for strange pups (McCarthy et al., 1986 [DOI](#)). Unpaired male mice's aggression towards pups was accompanied by changes in the activity of the vomeronasal neurons, and as males were paired with females and lived together, the activity of these neurons decreased, accompanied by a shift from infanticide to pup care behavior (Nakahara et al., 2020 [DOI](#)). The OTR expresses throughout the vomeronasal epithelium that allows OT to potentially inhibit infanticide by reducing vomeronasal activity. The previous study found that intraperitoneal injection of OT reduced the activity of the vomeronasal, and further validated that OT modulates the activity of vomeronasal neurons by acting on the sensory epithelium to produce behavioral changes by intraperitoneal injection of an OTR antagonist that cannot cross the blood-brain barrier (Nakahara et al., 2020 [DOI](#)). It has recently been shown that peripheral OT was able to cross the blood-brain barrier into the central nervous system to act (Yamamoto et al., 2019 [DOI](#)), meaning that delivery of OT via the periphery may have increased central OT levels and thus exerted an effect. A recent study demonstrated that the secretion of OT in the brain of male mice with no pairing experience facilitated the performance of pup care behaviors and inhibited infanticide (Inada et al., 2022 [DOI](#)), which also supported the results of the present study. Similar to rodents and non-human primates, there is evidence to suggest that OT contributes to paternal care (Feldman & Bakermans-Kranenburg, 2017 [DOI](#)). Fathers with partners have higher plasma OT levels than non-fathers without partners (Mascaro, Hackett, & Rilling, 2014 [DOI](#)). Intranasal OT treatment increases fathers' play, touch, and social interaction with their children (Weisman, Zagoory-Sharon, & Feldman, 2012 [DOI](#)). Our result provide possible application of OT in reduction of abnormality in pup-direct behavior associated with psychological diseases in human such as depression and psychosis (Milia & Noonan, 2022 [DOI](#); Naviaux, Janne, & Gourdin, 2020 [DOI](#)) and increases of well being of wild life.

In summary, these results indicate that the PVN to mPFC OT neural projection is involved in the regulation of pup care and infanticide behavior in virgin mandarin voles. These data provide new insights into the neural circuits underlying OT-mediated pup-directed behaviors.

Methods



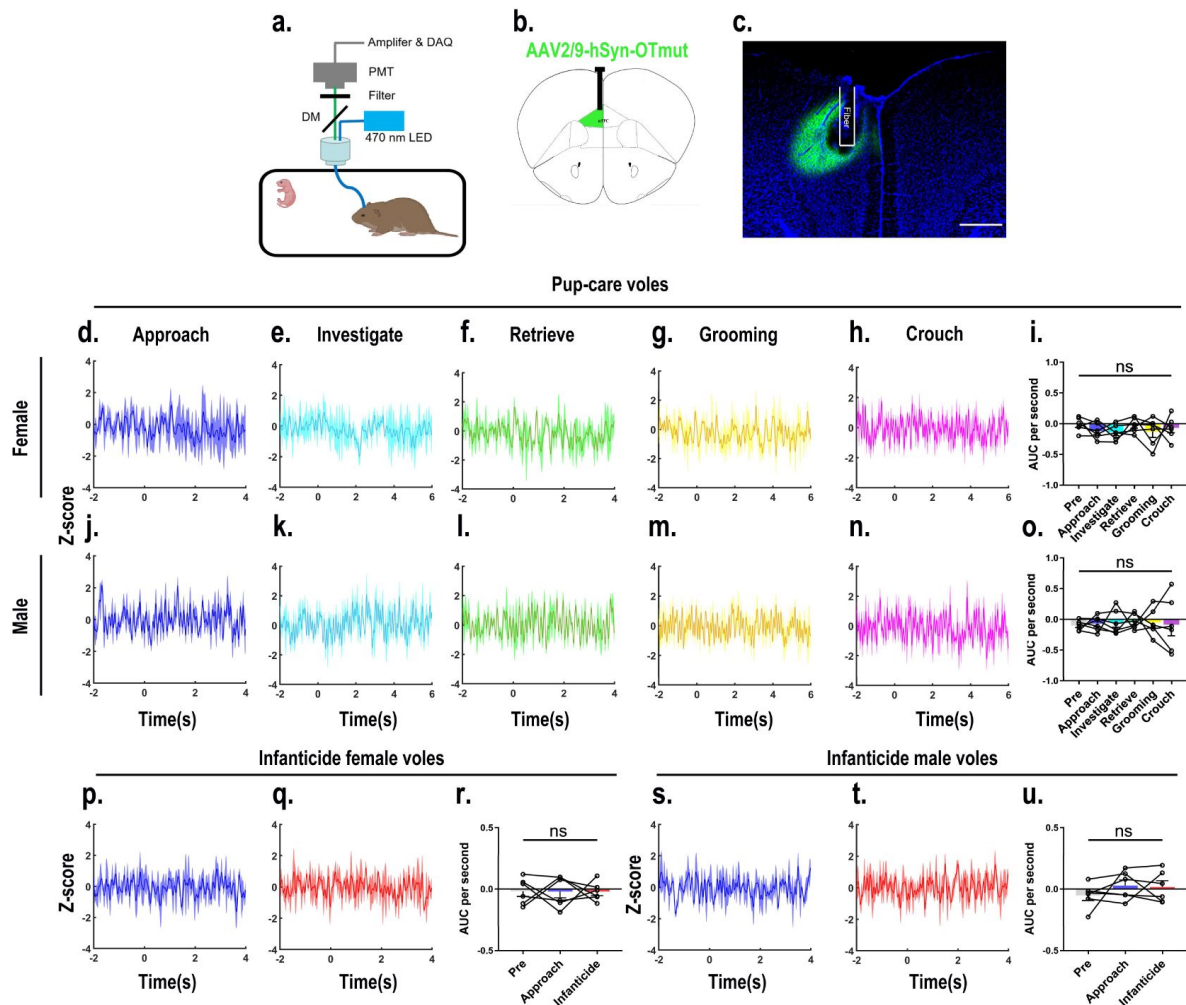
Supplementary Data Fig. 1

Light-induced c-Fos expression overlapping with mCherry, CHR2 or eNpHR

a, 473 nm light-induced c-Fos (green) expression overlapping with mCherry (red). Blue, DAPI. Objective: 20x. Scale bars, 50 μ m. b, 473 nm light-induced c-Fos (green) expression overlapping with CHR2 (red). Blue, DAPI. Objective: 20x. Scale bars, 50 μ m. c, 473 nm light induced more c-fos expression in CHR2, which indicated the effectiveness of CHR2 ($n = 3$). **** $P < 0.0001$. Independent-samples t-test. d, 589 nm light-induced c-Fos (green) expression overlapping with mCherry (red). Blue, DAPI. Objective: 20x. Scale bars, 50 μ m. e, 589 nm light-induced c-Fos (green) expression overlapping with eNpHR (red). Blue, DAPI. Objective: 20x. Scale bars, 50 μ m. f, 589 nm light induced less c-fos expression in eNpHR, which indicated the effectiveness of eNpHR ($n = 3$). ** $P < 0.01$. Independent-samples t-test. Error bars indicate SEM. For c,f, $n = 3$ per group, data are mean $\pm$ s.e.m. Statistical analysis was performed using independent-samples T test (c,f). ** $P < 0.01$, **** $P < 0.0001$. Details of the statistical analyses are as follows: c, $t(4) = -23.139$, $P = 0.000021$, f, $t(4) = 10.136$, $P = 0.001$.

Supplementary Data Fig. 1-source data 1

Statistical results of light-induced c-Fos expression overlapping with mCherry, CHR2 or eNpHR.



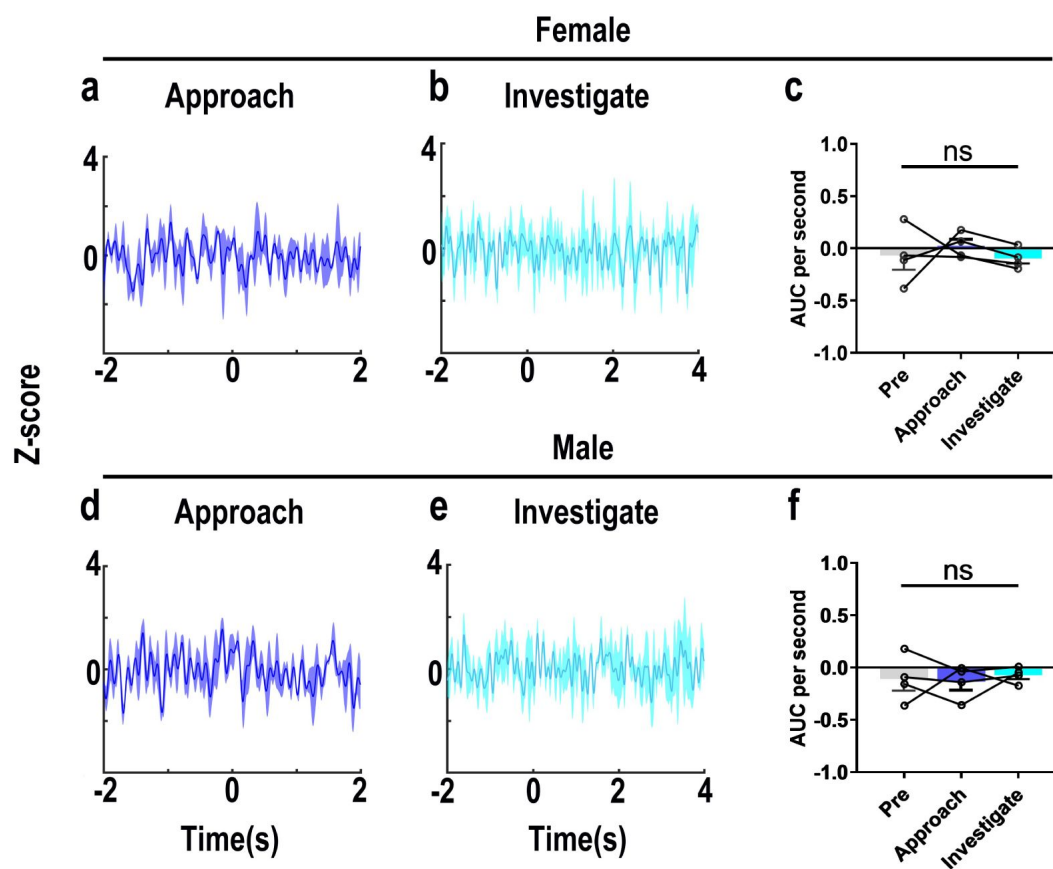
Supplementary Data Fig. 2

Recordings of OTmut sensor signals in the mPFC on OT release

a. Recording instrument settings. b. Illustrations of viral expression and optical fibre location. c. Representative histological image of OTmut sensor (green) and optical fibre locations. Blue, DAPI. Objective: 4x. Scale bars, 500 μ m. d-h,j-n, PETHs of z-score of OTmut sensor signals for the following pup-directed behaviors: approach (d,j), investigate (e,k), retrieve (f,l), grooming (g,m) and crouch (h,n). i,o, AUC per second of z-scores for pup-care female (n = 6, i) and male (n = 6, o) voles across various pup-directed behaviors. p,q,s,t, PETHs of z-score of OTmut sensor signals for approach and infanticide in infanticide voles. r,u, AUC per second of z-score of OTmut sensor signals for pre-pup exposure, approach and infanticide in infanticide female (n = 6, r) and male (n = 6, u) voles. Error bars indicate SEM. For i,o,r,u, n = 6 per group, data are mean $\pm$ s.e.m. Statistical analysis was performed using one-way rmANOVA (i,o,r and u). Details of the statistical analyses are as follows: i, $F(5, 25) = 0.680$, $P = 0.643$; o, $F(2, 25) = 0.074$, $P = 0.996$; r, $F(2, 10) = 0.002$, $P = 0.998$; u, $F(2, 10) = 0.665$, $P = 0.536$.

Supplementary Data Fig. 2-source data 1

Area under the curve per second for pre-pup exposure, approach, and infanticide behaviors in infanticide voles and area under the curve per second for pre-pup exposure, approach, investigation, retrieval, grooming, and crouching behaviors in pup-care voles.



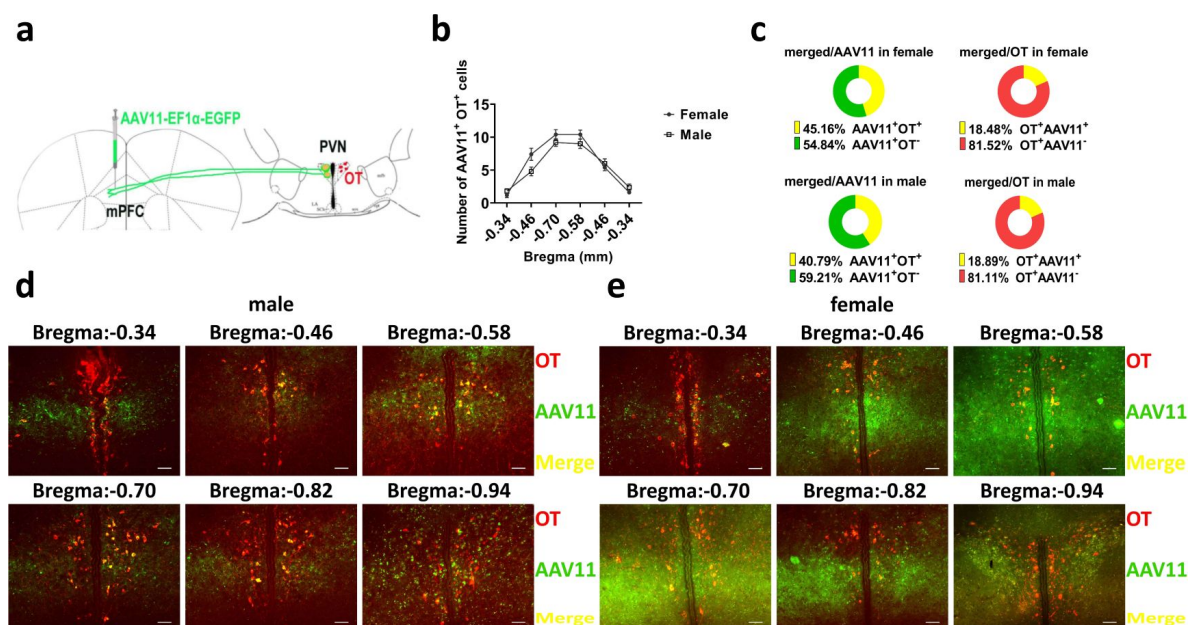
Supplementary Data Fig. 3

Recordings of OT1.0 sensor signals for investigating object

a,b,d,e, Post-event histograms (PEHs) of z-score of OT1.0 sensor for approaching and investigating object in female (a,b) and male (d,e) voles. c,f, AUC per second of z-scores for pre object, approaching and investigating behaviors in female (n = 4, c) and male (n = 4, f) voles. Error bars indicate SEM. For c,f, n = 4 per group, data are mean $\pm$ s.e.m. Statistical analysis was performed using one-way rmANOVA. Details of the statistical analyses are as follows: c, $F(1.016, 3.047) = 0.346$, $P = 0.601$; f, $F(2,6) = 0.183$, $P = 0.838$.

Supplementary Data Fig. 3-source data 1

Area under the curve per second for pre-object exposure, approach, and investigating in female and male voles.



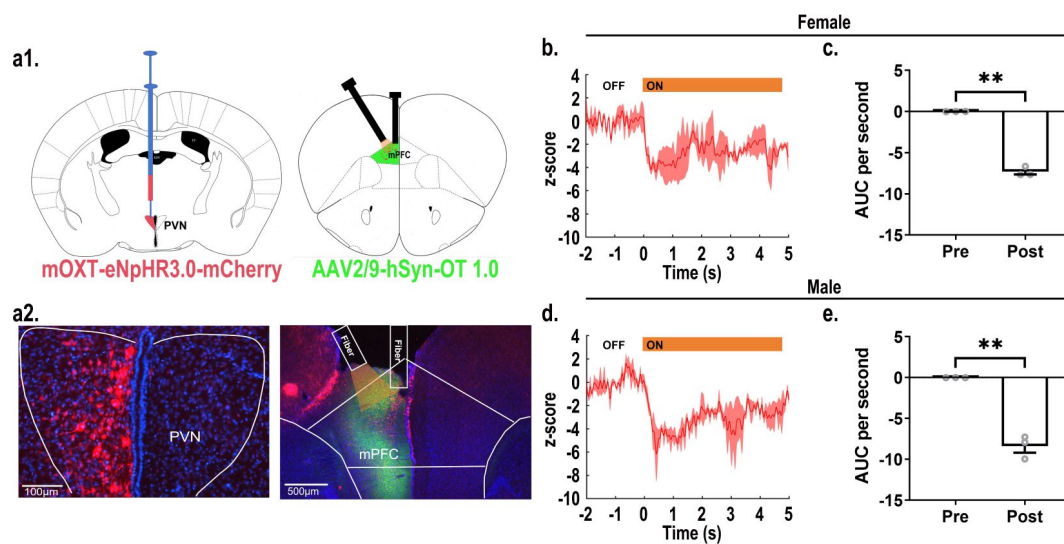
Supplementary Data Fig. 4

Number of PVN OT neurons projecting to mPFC

a. Schematic diagram of virus injection and immunohistological staining. b. Statistics of PVN to mPFC OT projection extent. c. Percentage of OT positive cells in PVN projecting to mPFC and percentage of OT cells in PVN that project to mPFC. d. e. Representative histologic images of male and female. Red, OT. Green, AAV11. Yellow, merged. Objective: 20x. Scale bars, 50 μ m.

Supplementary Data Fig. 4-source data 1

Counts of AAV11, OT and co-expressed cells.



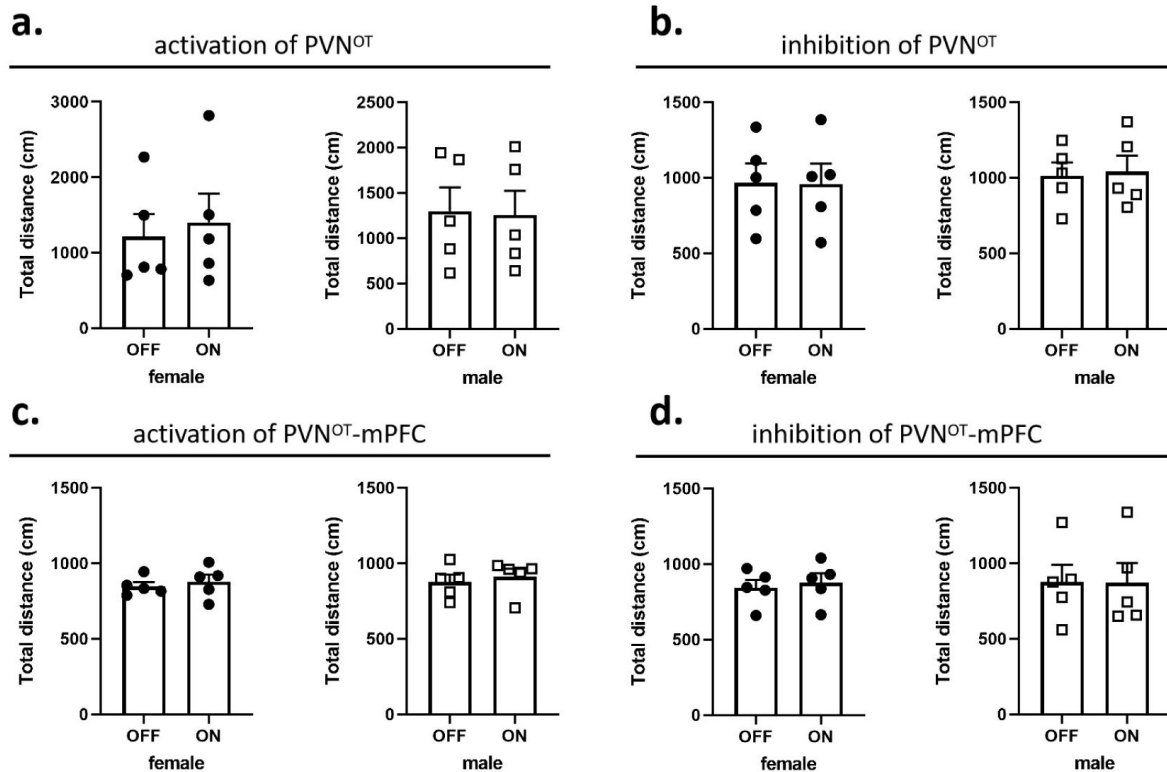
Supplementary Data Fig. 5

Recordings of OT release during optogenetic inhibition of OT fibers in the mPFC

a1. Schematic diagram of virus injection and optical fiber implanting. a2. Representative histological images of viruses infection and fibers placement. Left: Red: eNpHR3.0, Blue: DAPI, objective: 20x, scale bars: 100 μm; Right: Red: eNpHR3.0, Green: OT1.0 sensors, Blue: DAPI, objective: 4x, scale bars: 500 μm. b. Representative histogram of OT release in female at 589 nm light. c. AUC per second of OT1.0 sensors pre and post inhibition of optogenetics in females (n = 3). **P < 0.01, Paired-samples t-test. d. Representative histogram of OT release in male at 589 nm light. e. AUC per second of OT release pre and post optogenetic inhibition in males (n = 3). **P < 0.01, Paired-samples t-test.

Supplementary Data Fig. 5-source data 1

Statistics of AUC before and after optogenetic inhibition.



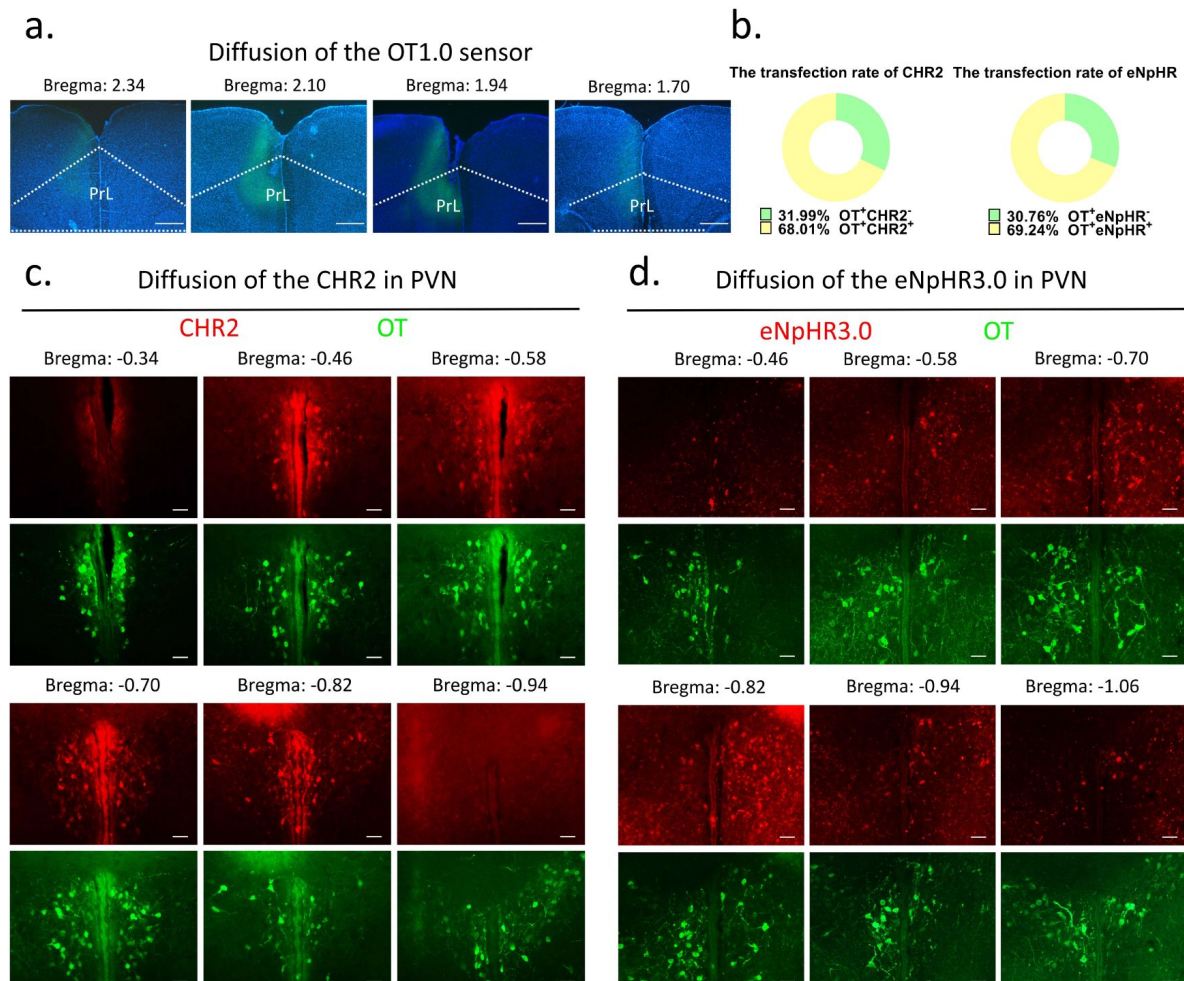
Supplementary Data Fig. 6

Effects of optogenetic manipulations on locomotion of the subjects

a. b. Total distance traveled by the subject before and after activation or inhibition of the PVN OT neurons. c. d. Total distance traveled by the subject before and after activation or inhibition of the PVN OT neurons projections to the mPFC.

Supplementary Data Fig. 6-source data 1

Statistics of distance traveled before and after optogenetic manipulation.



Supplementary Data Fig. 7

Diffusion of centrally injected agents

a. Diffusion of OT1.0 sensor in PrL. Green: OT1.0 sensors, Blue: DAPI, Objective: 4x. Scale bars: 500 μ m. b. The transfection rate of chr2 and eNpHR3.0 in OT cells. c. Diffusion of CHR2 in PVN. Green: OT, Red: CHR2, Objective: 20x. Scale bars: 50 μ m. d. Diffusion of eNpHR3.0 in PVN. Green: OT, Red: eNpHR3.0, Objective: 20x. Scale bars: 50 μ m.

Supplementary Data Fig. 7-source data 1

Counting of CHR2 and eNpHR3.0 virus-transfected OT cells.

Animals

Mandarin voles were captured from the wild in Henan, China. The experimental protocols were approved by the Animal Care and Use Committee of Shaanxi Normal University, and was conducted in accordance with the ethical principles of animal use and care of China. The virgin Mandarin voles (*Microtus mandarinus*) used in this study were F2 generations that were bred in Animal Center of Shaanxi Normal University and were kept at 24°C under a 12h light-dark cycle (lights on at 8 a.m.) with food and water provided ad libitum. Before the experiments, we expose the animals to pups, and subjects may exhibit pup care, infanticide, or neglect; we group subjects according to their behavioral responses to pups, and individuals who neglect pups are excluded. The stereotactic surgery was performed at the age of 8 weeks of voles. After surgery, voles were housed with their cage mates. Behavioral tests were carried out 3 weeks after surgery for animal recovery and the viral infection, and 1-5 days old pups were from other breeders. In case the pups were attacked, we removed them immediately to avoid unnecessary injuries, and injured pups were euthanized.

Viruses

AAV2/9-mOXT: Promoter-hCHR2(H134R)-mCherry-ER2-WPRE-pA (8.41×10^{12} µg/ml) and AAV2/9-mOXT: Promoter-mCherry-pA (1.21×10^{13} µg/ml) were purchased from Shanghai Taitool Bioscience LTD. AAV2/9-mOXT-eNpHR3.0-mCherry-WPRE-hGH-pA (2.27×10^{12} µg/ml) were purchased from BrainVTA (Wuhan, China) LTD. AAV2/9-hSyn-OT 1.0 (2.06×10^{12} µg/ml), AAV2/9-hSyn-OTmut (2.10×10^{12} µg/ml) and AAV11-EF1α-EGFP (5.00×10^{12} µg/ml) were purchased from Brain Case Biotechnology LTD. The details about the construction of CHR2 and mCherry viruses used in optogenetic manipulation can refer to a previous study in which they constructed an rAAV-expressing Venus from a 2.6 kb region upstream of OT exon 1, which is conserved in mammalian species (Knobloch et al., 2012 [DOI](#)). The details about construction of the eNpHR 3.0 virus can refer to one study in which expression of the vector is driven by the mouse OXT promoter, a 1kb promoter upstream of exon 1 of the OXT gene, which has been shown to induce cell type-specific expression in OXT cells (Peñagarikano et al., 2015 [DOI](#)). Details about the construction of OT 1.0 sensor can be referred to the research of Professor Li's group (Qian et al., 2023 [DOI](#)). All viruses were dispensed and stored at -80°C.

Immunohistochemistry

After behavioral tests, serial brain sections were harvested for histological analysis. Anesthetized voles were perfused with 40ml of PBS and 20ml of 4% paraformaldehyde. After perfusion, brains were excised and post-fixed by immersion in 4% paraformaldehyde overnight at 4°C. Brains were dehydrated in 20% and then 30% sucrose for 24h respectively before they were embedded in OCT and cryosectioned into 40 µm slices. Brain slices were rinsed with PBS (10 min) and PBST (PBS and 0.1% Triton X-100, 20 min), blocked in ready-to-use goat serum (Boster, AR0009) for 30 minutes at room temperature (RT), and then incubated overnight at 4°C with primary antibody in PBST. The primary antibodies used were: mouse anti-OT (1:7000, Millipore, MAB5296), rabbit anti-c-Fos (1:1000, Abcam, ab190289). Following PBST washing (3×5 min), sections were incubated with secondary antibodies in PBST for 2h (RT) and stained with DAPI, and then washed once more with PBS (3×5 min). The secondary antibodies used were: goat anti-mouse Alexa Fluor 488 (1:200, Jackson ImmunoResearch, 115-545-062), goat anti-rabbit Alexa Fluor 488 (1:200, Jackson ImmunoResearch, 111-545-003), goat anti-rabbit TRITC (1:200, Jackson ImmunoResearch, 111-025-003) and ready-to-use DAPI staining solution (Boster, AR1177). Finally, the brain slices were sealed with an anti-fluorescent attenuation sealer.

Images were captured with a fluorescent microscope (Nikon) to confirm the viral expression, placements of optic fiber and viruses, and also the number of c-Fos, OT and virus positive cells. To analyze the activity of OT neurons (co-expression of c-Fos and OT) among different behaviors and the specificity of viral expression (co-expression of viruses and OT) in the PVN brain region, Brain

slices of 40 μm were collected consecutively on 4 slides, each slide had 6 brain slices spaced 160 μm apart from each other, and counting was performed on one of the slides. Positive cells in the PVN were manually counted based on the Allen Mouse Brain Atlas and our previous studies.

Stereotaxic Surgery

For optogenetic manipulation experiments, CHR2, eNpHR and mCherry-expressing control virus were stereotactically injected into the PVN (AP: -0.4 mm, ML: 0.2 mm, DV: 5.3mm) bilaterally through a Hamilton needle using nanoinjector (Reward Life Technology, KDS LEGATO 130) at 100 nl/min. For optogenetic manipulation, optic fibers (2.5 mm O.D., Reward Life Technology, China) with an appropriate fiber length (PVN: 6 mm, mPFC: 3 mm) were implanted $\sim 100\ \mu\text{m}$ above the PVN and mPFC (AP: 2.2 mm ML: 0.9, DV: 1.89 with a 20° angle lateral to middle) and secured with dental cement (Changshu ShangChi Dental Materials Co., Ltd., 202005). For fiber optometry experiment, optic fibers were inserted $\sim 100\ \mu\text{m}$ above the mPFC (AP: 2.2 mm ML: 0.3, DV: 1.8) after injecting the OT1.0 sensor viruses. The stereotaxic coordinates were determined from three-dimensional brain atlas (Chan, Kovacevic, Ho, Henkelman, & Henderson, 2007 [DOI](#)) and the adjusted coordinates in our lab. The stereotaxic coordinates were determined by the Allen Mouse Brain Atlas and laboratory-corrected data used for voles. Individuals with appropriate viral expression and optical fiber embedding location were included in the statistical analysis, otherwise excluded. The diffusion of central optogenetic viruses and OT1.0 sensors are shown in the supplemental figure (Supplementary Data Fig. 7 [DOI](#)).

Optogenetic

Test animals were injected with 300 nL AAV2/9-mOXT: Promoter-hCHR2(H134R)-mCherry-ER2-WPRE-pA or rAAV-mOXT-eNpHR3.0-mCherry-WPRE-hGH-pA bilaterally into the PVN at 100 nl/min. Control animals were injected with AAV2/9-mOXT: Promoter-mCherry-pA in the same condition. Two weeks later, an optic fiber was implanted $\sim 100\ \mu\text{m}$ above the PVN and mPFC bilaterally and was secured using dental cement. After surgery, animals were housed with their cage mates as before. Before the behavioral test, each implanted fiber was connected to a light laser (NewdoonInc., Aurora 300, 473 nm for activation, 589 nm for inhibition) with a 400 μm patch cord and then we introduced the test animal into center of test arena. After the test animal acclimated to the arena for at least 20 min and settled down, we placed a pup in the farthest corner away from the test animal and immediately began recording and applying light stimulation.

In pup-directed pup care behavioral test, light stimulation lasted for 11 min. Parameters used in optogenetic manipulation of PVN OT neurons were $\sim 3\ \text{mW}$, 20 Hz, 20 ms, 8 s ON and 2 s OFF and parameters used in optogenetic manipulation of PVN OT neurons projecting to mPFC were $\sim 10\ \text{mW}$, 20 Hz, 20 ms, 8 s ON and 2 s OFF to cover the entire interaction. In the infanticide behavioral test, the stimulation lasted until the pup was removed. Each vole was tested twice successively, more than 30 min apart, once with the stimulation OFF, once ON. The effect of optogenetic manipulation induced locomotion on behavioral responses to pups was excluded by recording the total distance traveled by voles without and with light stimulation for 5 minutes, respectively (Supplementary Data Fig. 6 [DOI](#)).

To confirm that Chr2 or eNpHR3.0 stimulation indeed induced neural activation or inhibition, we used light to stimulate the brain through an optical fiber when voles were alone in their home cage, and subsequently determined neural activation or inhibition by c-Fos staining one and a half hours after light stimulation.

Fiber photometry

To record the fluorescence signals of the OT1.0 sensor during various pup-directed behaviors. Virgin voles were anesthetized with 1-2% isoflurane and immobilized on a stereotaxic device (RWD, China). Then, 300 nl AAV9-hSyn-OT1.0 or AAV9-hSyn-OTmut virus was injected into the left

side of mPFC (AP: 2.2 mm ML: 0.3, DV: 1.8). After the injection, a 200 μ m optical fiber was implanted $\sim$ 100 μ m above the injection site and fixed with dental cement. After 2 weeks of recovery, the optic fiber was connected to the fiber photometry system (QAXK-FPS-LED, ThinkerTech, Nanjing, China) through a patch cable. This system can automatically exclude motion artifacts by simultaneously recording signals stimulated by a 405 nm light source. To avoid bleaching the sensor, the 470 nm laser power at the tip of the fiber was adjusted to 50 μ W. Voles were placed in test cages and allowed to move freely for at least 20 minutes to acclimate to the environment. Then, a pup was placed in the cage at a distance from the testing vole. If the vole ignored the pup completely or attacked the pup, gently removed the pup and introduced another pup $\sim$ 60 seconds later to stimulate more interaction. This process was repeated 3-6 times, and then the vole exhibiting pup care behavior was allowed to freely interact with the last introduced pup until the vole crouched over the pup for more than 10 s, after which the pups were removed for about 60 s and reintroduced. This latter process was repeated 3-4 times. After the pup test, we subsequently placed an object (a vole-sized plastic toy) into the cage and recorded 4 voles that investigated the object 3-6 times.

Videos were recorded with screen-recoding software to synchronize the OT1.0 fluorescence signals and pup-directed behaviors. Fluorescence signals were recorded into MATLAB mat files and analyzed with customized MATLAB code. Data were matched to a variety of behaviors towards pups based on individual trials. The change in signal was displayed as z-scored $\Delta F/F$, which was measured by $(V_{\text{signal}} - \text{mean}(V_{\text{basal}}))/\text{std}(V_{\text{basal}})$. The V_{signal} and V_{basal} refer to the recorded values at each time point and the recorded values during the baseline period before the stimuli. The AUC (area under the curve) was calculated based on z-scored $\Delta F/F$ matching the duration of the behavior, and the AUC per second was used to compare the different fluorescence signals of behaviors and the baseline.

For the combination of optogenetic inhibition and fiber optometry experiment, optogenetic virus and OT1.0 sensor were injected as described above, optic fibers were inserted above the mPFC (OT1.0 fibers: AP: 2.2 mm ML: 0.2, DV: 1.7; optogenetic fibers: AP: 2.2 mm ML: 2.0, DV: 2.4 with a 45° angle lateral to middle). The signals of OT1.0 sensor were recorded while neurons were optogenetically inhibited.

Behavioral paradigm and analysis

Animal behaviors in optogenetic experiments were recorded by a camera from the side of a transparent cage. ‘Approach pup’ was defined as the testing vole faced and walked right up to pup, and the latency to approach was the period from the time the pup was placed in the cage until the vole began to approach the pup. ‘Investigate pup’ was defined as the vole’s nose came into close contact with any part of the pup’s body. ‘Attack pup’ was defined as the vole attacked or bit a pup which can be recognized by the wound, and the latency to attack was the period from the time the pup was placed in the cage until the vole launched an attack. ‘Retrieve pup’ was defined as from the time a vole picked up a pup using its jaws to the time it dropped the pup at or around the nest, and the latency of retrieval was the time between the pup was put in the cage and the time the vole picked up the pup in its jaws. ‘Groom pup’ was defined as a vole combed the pup’s body surface with its muzzle, accompanied by a rhythmic up-and-down bobbing of the vole’s head and displacement of the pup. ‘Crouch’ was defined as the vole squatted quietly over the pup with no apparent movement. Pup-directed behaviors in optogenetic experiments were scored and analyzed by JWatcher.

OT treatments

Test virgin voles were acclimatized in their cages for 20 min before being injected intraperitoneally with 0.9% NaCl (1 ml/kg) and pups were introduced 30 min later. The behavioral responses were recorded from the side using a video camera. Thirty mins after the first record,

voles were re-injected intraperitoneally OT 1 mg/kg (Leuner, Caponiti, & Gould, 2012 [DOI](#)) (Bachem, 50-56-6), and pups were introduced 30 min later and behavioral responses were recorded. All behaviors were scored and analyzed using JWatcher.

Statistics

Parametric tests were used to analyze normally distributed data, and nonparametric tests were used for data that is not normally distributed according to Kolmogorov–Smirnov tests. Independent samples t-tests (two-tailed) were performed to assess number of OT, c-fos and merge rate of OT & c-fos during different pup-directed behaviors (Pup-care vs. Infanticide) and the number of c-fos-IR positive neurons (mCherry vs. CHR2; mCherry vs. eNpHR 3.0). The behavioral changes following optogenetic activation and inhibition (factors: treatment × stimulus) of PVN and mPFC were analyzed by two-way repeated-measures ANOVA. One way ANOVA were used to analyze the AUC changes recorded by the OT sensors in different behaviors. Paired-samples t-test (two-tailed) and Wilcoxin signed ranks test were used to analyze changes in pup-directed behaviors before and after intraperitoneal injection of OT. The Pearson chi-square test was used to compare the difference in the number of infanticide voles between the saline and OT groups. All data were presented as mean ± SEM, statistical analyses of data were performed using MATLAB and SPSS 22.0 software.

Conflict of interest

The authors declare no conflict of interest.

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Reviewer #1 (Public review):

Summary:

This important study investigated the role of oxytocin (OT) neurons in the paraventricular nucleus (PVN) and their projections to the medial prefrontal cortex (mPFC) in regulating pup care and infanticide behaviors in mandarin voles. The researchers used techniques like immunofluorescence, optogenetics, OT sensors, and peripheral OT administration. Activating OT neurons in the PVN reduced the time it took pup-caring male voles to approach and retrieve pups, facilitating pup care behavior. However, this activation had no effect on

females. Interestingly, this same PVN OT neuron activation also reduced the time for both male and female infanticidal voles to approach and attack pups, suggesting PVN OT neuron activity can promote pup care while inhibiting infanticide behavior. Inhibition of these neurons promoted infanticide. Stimulating PVN->mPFC OT projections facilitated pup care in males and in infanticide prone voles, activation of these terminals prolonged latency to approach and attack. Inhibition of PVN->mPFC OT projections promoted infanticide. Peripheral OT administration increased pup care in males and reduced infanticide in both sexes. However, some results differed in females, suggesting other mechanisms may regulate female pup care.

Strengths:

This multi-faceted approach provides converging evidence and strengthens the conclusions drawn from the study and make them very convincing. Additionally, the study examines both pup care and infanticide behaviors, offering insights into the mechanisms underlying these contrasting behaviors. The inclusion of both male and female voles allows for the exploration of potential sex differences in the regulation of pup-directed behaviors. The peripheral OT administration experiments also provide valuable information for potential clinical applications and wildlife management strategies.

Weaknesses:

While the study presents exciting findings, there are several weaknesses. The sample sizes used in some experiments, such as the Fos study and optogenetic manipulations, appear to be small, which may limit the statistical power and generalizability of the results.

There is potential effect of manipulating OT neurons on the release of other neurotransmitters (or the influence of other neurochemicals or brain regions) on pup-directed behaviors, especially in females, are not fully explored. Additionally, it is unclear whether back-propagation of action potentials during optogenetic manipulations causes the same behavioral effect as direct stimulation of PVN OT cells. However, the authors now discuss these possibilities. It is also uncertain whether more OT neurons were manipulated in females compared to males. All other comments have been addressed by the authors.

<https://doi.org/10.7554/eLife.96543.2.sa3>

Reviewer #2 (Public review):

Summary:

This series of experiments studied the involvement of PVN OT neurons and their projection to the mPFC in pup-care and attack behavior in virgin male and female Mandarin voles. Using Fos visualization, optogenetics, fiber photometry and IP injection of OT the results converge on OT regulating caregiving and attacks on pups. Some sex differences were found in the effects of the manipulations.

Strengths:

Major strengths are the modern multi-method approach and including both sexes of Mandarin vole in every experiment.

Weaknesses:

The few weaknesses include 1) Some experiments' groups have small sample sizes (4-5 animals) which may render some results difficult for others to replicate when different extraneous variables are likely to be present, and 2) the authors discuss PVN OT cell stimulation findings seen in other rodents so the work seems less conceptually novel. Overall,

the findings add to the knowledge about OT regulation of pup-directed behavior in male and female rodents, especially the PVN-mPFC OT

<https://doi.org/10.7554/eLife.96543.2.sa2>

Reviewer #3 (Public review):

Summary:

Here Li et al. examine pup-directed behavior in virgin Mandarin voles. Some males and females tend towards infanticide, others tend towards pup care. c-Fos staining showed more oxytocin cells activated in the paraventricular nucleus (PVN) of the hypothalamus in animals expressing pup care behaviors than in infanticidal animals. Optogenetic stimulation of PVN oxytocin neurons (with an oxytocin-specific virus to express the opsin transgene) increased pup-care, or in infanticidal voles increased latency towards approach and attack. Suppressing activity of PVN oxytocin neurons promoted infanticide. Use of a recent oxytocin GRAB sensor (OT1.0) showed changes in medial prefrontal cortex (mPFC) signals as measured with photometry in both sexes. Activating mPFC oxytocin projections increased latency to approach and attack in infanticidal females and males (similar to the effects of peripheral oxytocin injections), whereas in pup caring animals only males showed a decrease to approach. Inhibiting these projections increased infanticidal behaviors in both females and males, and no effect on pup caretaking.

Strengths:

Adopting these methods for Mandarin voles is an impressive accomplishment, especially the valuable data provided by the oxytocin GRAB sensor. This is a major achievement and helps promote systems neuroscience in voles.

The authors have done a good job responding to the comments on their preprint. I'd ask them to check their z-scored values, as the mean of a z-scored value should be 0 over time. Also I'm not sure I agree that the fiber photometry system "can automatically exclude effects of motion artifacts"; yes that is a function of imaging at a different wavelength but that process is also prone to error and imperfect.

<https://doi.org/10.7554/eLife.96543.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

This important study investigated the role of oxytocin (OT) neurons in the paraventricular nucleus (PVN) and their projections to the medial prefrontal cortex (mPFC) in regulating pup care and infanticide behaviors in mandarin voles. The researchers used techniques like immunofluorescence, optogenetics, OT sensors, and peripheral OT administration. Activating OT neurons in the PVN reduced the time it took pup-caring male voles to approach and retrieve pups, facilitating pup-care behavior. However, this activation had no effect on females. Interestingly, this same PVN OT neuron activation also reduced the time for both male and female infanticidal voles to approach and attack pups,

suggesting PVN OT neuron activity can promote pup care while inhibiting infanticide behavior. Inhibition of these neurons promoted infanticide. Stimulating PVN->mPFC OT projections facilitated pup care in males and in infanticide-prone voles, activation of these terminals prolonged latency to approach and attack. Inhibition of PVN->mPFC OT projections promoted infanticide. Peripheral OT administration increased pup care in males and reduced infanticide in both sexes. However, some results differed in females, suggesting other mechanisms may regulate female pup care.

Strengths:

This multi-faceted approach provides converging evidence, strengthens the conclusions drawn from the study, and makes them very convincing. Additionally, the study examines both pup care and infanticide behaviors, offering insights into the mechanisms underlying these contrasting behaviors. The inclusion of both male and female voles allows for the exploration of potential sex differences in the regulation of pup-directed behaviors. The peripheral OT administration experiments also provide valuable information for potential clinical applications and wildlife management strategies.

Weaknesses:

While the study presents exciting findings, there are several weaknesses that should be addressed. The sample sizes used in some experiments, such as the Fos study and optogenetic manipulations, appear to be small, which may limit the statistical power and generalizability of the results. Effect sizes are not reported, making it difficult to evaluate the practical significance of the findings. The imaging parameters and analysis details for the Fos study are not clearly described, hindering the interpretation of these results (i.e., was the entire PVN counted?). Also, does the Fos colocalization align with previous studies that look at PVN Fos and maternal/ paternal care? Additionally, the study lacks electrophysiological data to support the optogenetic findings, which could provide insights into the neural mechanisms underlying the observed behaviors.

In some previous studies (He et al., 2019; Mei, Yan, Yin, Sullivan, & Lin, 2023), the sample size in morphological studies is also small and may be representative. We agree with reviewer's opinion that results from larger sample size may be more statistically powerful and generalizable. We will pay attention to this issue in the future study. As reviewer suggested, we have added effect size both in the source data and in the main text, including d , η^2 and odds ratio. We have added the objective magnification used in the figure legend. The imaging parameters and analysis details for the Fos study have also been added in the revised manuscript. Brain slices of 40 μm thick were collected consecutively on 4 slides, each slide had 6 brain slices spaced 160 μm apart from each other. PVN area were determined based on the Allen Mouse Brain Atlas and our previous study, and Fos, OT and merged positive neurons were counted. Our result about Fos and OT colocalization is consistent with previous study. In a previous study on virgin male prairie voles, OT and Fos colabeled neurons in the PVN increased after exposure to conspecific pups and experiencing paternal care (Kenkel et al., 2012). In another study of prairie voles, OT and c-fos colabeled neurons in PVN significantly increased after becoming parents which may be due to a shift from virgin to parents (Kelly, Hiura, Saunders, & Ophir, 2017). To support the optogenetic findings, we used c-Fos expression as a marker of neuron activity and revealed significant increases/decreases of c-Fos positive neurons induced by optogenetic activation/inhibition (Supplementary Data Fig. 1), and additionally we found that optogenetic inhibition of OT neurons reduced levels of OT release using OT1.0 sensors. Based on these two experiments, we verified that optogenetic manipulation in the present study is validate and results of optogenetic experiment are reliable (Supplementary Data Fig. 5).

The study has several limitations that warrant further discussion. Firstly, the potential effects of manipulating OT neurons on the release of other neurotransmitters (or the influence of other neurochemicals or brain regions) on pup-directed behaviors, especially in females, are not fully explored. Additionally, it is unclear whether back-propagation of action potentials during optogenetic manipulations causes the same behavioral effect as direct stimulation of PVN OT cells. Moreover, the authors do not address whether the observed changes in behavior could be explained by overall increases or decreases in locomotor activity.

We agree with reviewer's suggestion that several limitations should be discussed. Although we used a virus strategy to specifically activate or inhibit PVN OT neurons, other neurochemical may also be released during optogenetic manipulations because OT neurons may also release other neurochemicals. In one of our previous studies, activation of the OT neuron projections from the PVN to the VTA as well as to the Nac brain also altered pup-directed behaviors, which may also be accompanied by dopamine release (He et al., 2021). In addition, backpropagation of action potentials during optogenetic manipulations may also causes the same behavioral effect as direct stimulation of PVN OT cells. These effects on pup-directed behaviors should also be investigated further in the future study. For the optogenetics experiments, we have referred to some of the previous research (Mei et al., 2023; Murugan et al., 2017), and in our study we have also carried out the verification of the reliability of the methods. To exclude effects of locomotor activity on pup directed behaviors, we also investigated effect of optogenetic manipulations on the locomotor activity of experimental animals and found that optogenetic manipulation did not change levels of locomotor activity (Supplementary Data Fig. 6).

The authors do not specify the percentage of PVN->mPFC neurons labeled that were OT-positive, nor do they directly compare the sexes in their behavioral analysis (or if they did, it is not clear statistically). While the authors propose that the sex difference in pup-directed behaviors is due to females having greater OT expression, they do not provide evidence to support this claim from their labeling data. It is also uncertain whether more OT neurons were manipulated in females compared to males. The study could benefit from a more comprehensive discussion of other factors that could influence the neural circuit under investigation, especially in females.

AAV11-Ef1a-EGFP virus can infect fibers and retrogradely reach to cell body, thus this virus can be used to retrogradely trace neurons. We injected this virus (green, AAV11-Ef1a-EGFP) in the mPFC and observed virus infected and OT (red) positive neuron in the PVN (Yellow), and we also counted the OT neurons that project from PVN to mPFC and found that approximately 45.16% and 40.79% of cells projecting from PVN to the mPFC were OT-positive, and approximately 18.48% and 18.89% of OT cells in the PVN projected to the mPFC in females and males, respectively (Supplementary Data Fig. 4). In addition, as reviewers suggested, we compared the numbers of OT neurons, activated OT neurons (OT and Fos double-labeled neurons) and level of OT release between males and females. We found that females have more activated OT neurons (Figure1, d, g) and released higher levels of OT into the mPFC (Figure 4 d, e) than males. This part has been added in the result and discussion. We did not analyze whether more OT neurons were manipulated in females compared to males, which is indeed a limitation of this study that requires our attention.

As the reviewers suggested, we also discussed other factors that could influence the neural circuit under investigation. In addition to OT neurons, OTR neurons may also regulate behavioral responses to pups. In a study of virgin female mice, pup exposure was found to activate oxytocin and oxytocin receptor expressing neurons (Okabe et al., 2017). Other brain regions such as preoptic area (POA) may also be involved in parental behaviors. For example,

virgin female mice repeatedly exposed to pups showed shorter retrieval latencies and greater c-Fos expression in the preoptic area (POA), concentrations of OT in the POA were also significantly increased, and the facilitation of alloparental behavior by repeated exposure to pups occurred through the organization of the OT system (Okabe et al., 2017). A recent study suggests that OT of the PVN is involved in the care of pups by male voles (He et al., 2021). This study suggests that PVN to ventral tegmental area (VTA) OT projections as well as VTA to nucleus accumbens (NAc) DA projections are involved in the care of pups by male voles. Inhibition of OT projections from the PVN to the VTA reduces DA release in the NAc during licking and grooming of pups (He et al., 2021). The effects of these factors on pup-directed responses should also be considered in the future study.

Reviewer #2 (Public Review):

Summary:

This series of experiments studied the involvement of PVN OT neurons and their projection to the mPFC in pup-care and attack behavior in virgin male and female Mandarin voles. Using Fos visualization, optogenetics, fiber photometry, and IP injection of OT the results converge on OT regulating caregiving and attacks on pups. Some sex differences were found in the effects of the manipulations.

Strengths:

Major strengths are the modern multi-method approaches and involving both sexes of Mandarin vole in every experiment.

Weaknesses:

Weaknesses include the lack of some specific details in the methods that would help readers interpret the results. These include:

(1) No description of diffusion of centrally injected agents.

Thanks for your professional consideration. Individuals with appropriate viral expression and optical fiber implant location were included in the statistical analysis, otherwise excluded. For optogenetic experiments, the virus (AAV2/9-mOXT-hCHR2(H134R)-mCherry-ER2-WPRE-pA or rAAV-mOXT-eNpHR3.0-mCherry-WPRE-hGH-pA) was designed and constructed to only infect OT neurons, which limited the diffusion of the virus. For fiber photometric experiments, the OT1.0 sensor was largely able to restrict expression within the mPFC brain region, and additionally individuals with incorrect optical fiber embedding position were not included in the statistical analysis. The diffusion of central optogenetic viruses and OT1.0 sensors are shown in the supplemental figure (Supplementary Data Fig. 7).

(2) Whether all central targets were consistent across animals included in the data analyses. This includes that is not stated if the medial prelimbic mPFC target was in all optogenetic study animals as shown in Figure 4 and if that is the case, there is no discussion of that subregion's function compared to other mPFC subregions.

As shown in Figure 4 and in the schematic diagram of the optogenetic experiment, the central targets of virus infection and fiber location remain consistent in the data analysis, otherwise the data would be excluded. In the present study, viruses were injected into the prelimbic (PrL). The PrL and infralimbic (IL) regions of the mPFC play different roles in different social interaction contexts (Bravo-Rivera, Roman-Ortiz, Brignoni-Perez, Sotres-Bayon, & Quirk, 2014; Moscarello & LeDoux, 2013). A study has shown that the PrL region of the mPFC contributes to active avoidance in situations where conflict needs to be mitigated, but also contributes to the retention of conflict responses for reward (Capuzzo & Floresco, 2020). This may reveal

that the suppression of infanticide by PVN to mPFC OT projections is a behavioral consequence of active conflict avoidance. In a study on pain in rats, OT neurons projections from the PVN to the PrL were found to increase the responsiveness of cell populations in the PrL, suggesting that OT may act by altering the local excitation-inhibition (E/I) balance in the PrL (Liu et al., 2023). A study on anxiety-related behaviors in male rats suggests that the anxiolytic effects of OT in the mPFC are PrL-specific but not infralimbic or anterior cingulate and that this is achieved primarily through the engagement of GABAergic neurons, which ultimately modulate downstream anxiety-related brain regions, including the amygdala (Sabihi, Dong, Maurer, Post, & Leuner, 2017). This finding may provide possible downstream pathways for further research.

(3) How groups of pup-care and infanticidal animals were created since there was no obvious pretest mentioned so perhaps there was the testing of a large number of animals until getting enough subjects in each group.

Before the experiments, we exposed the animals to pups, and subjects may exhibit pup care, infanticide, or neglect; we grouped subjects according to their behavioral responses to pups, and individuals who neglected pups were excluded.

(4) The apparent use of a 20-minute baseline data collection period for photometry that started right after the animals were stressed from handling and placement in the novel testing chamber.

In fiber photometric experiments, all experimental animals were required to acclimatize to the environment for at least 20 minutes prior to the experiment as described in the Methods section. The time 0 in Fig. 4 represents the point in time when a behavior or a segment of behavior started and is not the actual time 0 at which the test was started.

(5) A weakness in the results reporting is that it's unclear what statistics are reported (2 x 2 ANOVA main effect of interaction results, t-test results) and that the degrees of freedom expected for the 2 X 2 ANOVAs in some cases don't appear to match the numbers of subjects shown in the graphs; including sample sizes in each group would be helpful because the graph panels are very small and data points overlap.

Thanks for your suggestion. We displayed analysis methods for the data statistics and the sample sizes for each group of experiments in the figure legends.

The additional context that could help readers of this study is that the authors overlook some important mPFC and pup caregiving and infanticide studies in the introduction which would help put this work in better context in terms of what is known about the mPFC and these behaviors. These previous studies include Febo et al., 2010; Febo 2012; Peirera and Morrell, 2011 and 2020; and a very relevant study by Alsina-Llanes and Olazábal, 2021 on mPFC lesions and infanticide in virgin male and female mice. The introduction states that nothing is known about the mPFC and infanticide. In the introduction and discussion, stating the species and sex of the animals tested in all the previous studies mentioned would be useful. The authors also discuss PVN OT cell stimulation findings seen in other rodents, so the work seems less conceptually novel. Overall, the findings add to the knowledge about OT regulation of pup-directed behavior in male and female rodents, especially the PVN-mPFC OT projection.

We appreciate you very much to provide so many valuable references. We have cited them in the introduction and discussion. We agree with the reviewer's opinion that nothing is known about the mPFC and infanticide is incorrect. It should be whether mPFC OT projections are involved in paternal cares and infanticide remains unclear. A study in mother rats indicated

that inactivation or inhibition of neuronal activity in the mPFC largely reduced pup retrieval and grouping (Febo, Felix-Ortiz, & Johnson, 2010). In a subsequent study on firing patterns in the mPFC of mother rats suggested that sensory-motor processing occurs in the mPFC that may affect decision making of maternal care to their pups (Febo, 2012). In a study on new mother rats examining different regions of the mPFC (anterior cingulate (Cg1), PrL, IL), they identified a involvement of the IL cortex in biased preference decision-making in favour of the offspring (Pereira & Morrell, 2020). A study on maternal motivation in rats suggests that in the early postpartum period, the IL and Cg1 subregion in mPFC, are the motivating circuits for pup-specific biases (Pereira & Morrell, 2011), while the PrL subregion, are recruited and contribute to the expression of maternal behaviors in the late postpartum period (Pereira & Morrell, 2011).

Reviewer #3 (Public Review):

Summary:

Here Li et al. examine pup-directed behavior in virgin Mandarin voles. Some males and females tend towards infanticide, others tend towards pup care. c-Fos staining showed more oxytocin cells activated in the paraventricular nucleus (PVN) of the hypothalamus in animals expressing pup care behaviors than in infanticidal animals. Optogenetic stimulation of PVN oxytocin neurons (with an oxytocin-specific virus to express the opsin transgene) increased pup-care, or in infanticidal voles increased latency towards approach and attack.

Suppressing the activity of PVN oxytocin neurons promoted infanticide. The use of a recent oxytocin GRAB sensor (OT1.0) showed changes in medial prefrontal cortex (mPFC) signals as measured with photometry in both sexes. Activating mPFC oxytocin projections increased latency to approach and attack in infanticidal females and males (similar to the effects of peripheral oxytocin injections), whereas in pup-caring animals only males showed a decrease in approach. Inhibiting these projections increased infanticidal behaviors in both females and males and had no effect on pup caretaking.

Strengths:

Adopting these methods for Mandarin voles is an impressive accomplishment, especially the valuable data provided by the oxytocin GRAB sensor. This is a major achievement and helps promote systems neuroscience in voles.

Weaknesses:

The study would be strengthened by an initial figure summarizing the behavioral phenotypes of voles expressing pup care vs infanticide: the percentages and behavioral scores of individual male and female nulliparous animals for the behaviors examined here. Do the authors have data about the housing or life history/experiences of these animals? How bimodal and robust are these behavioral tendencies in the population?

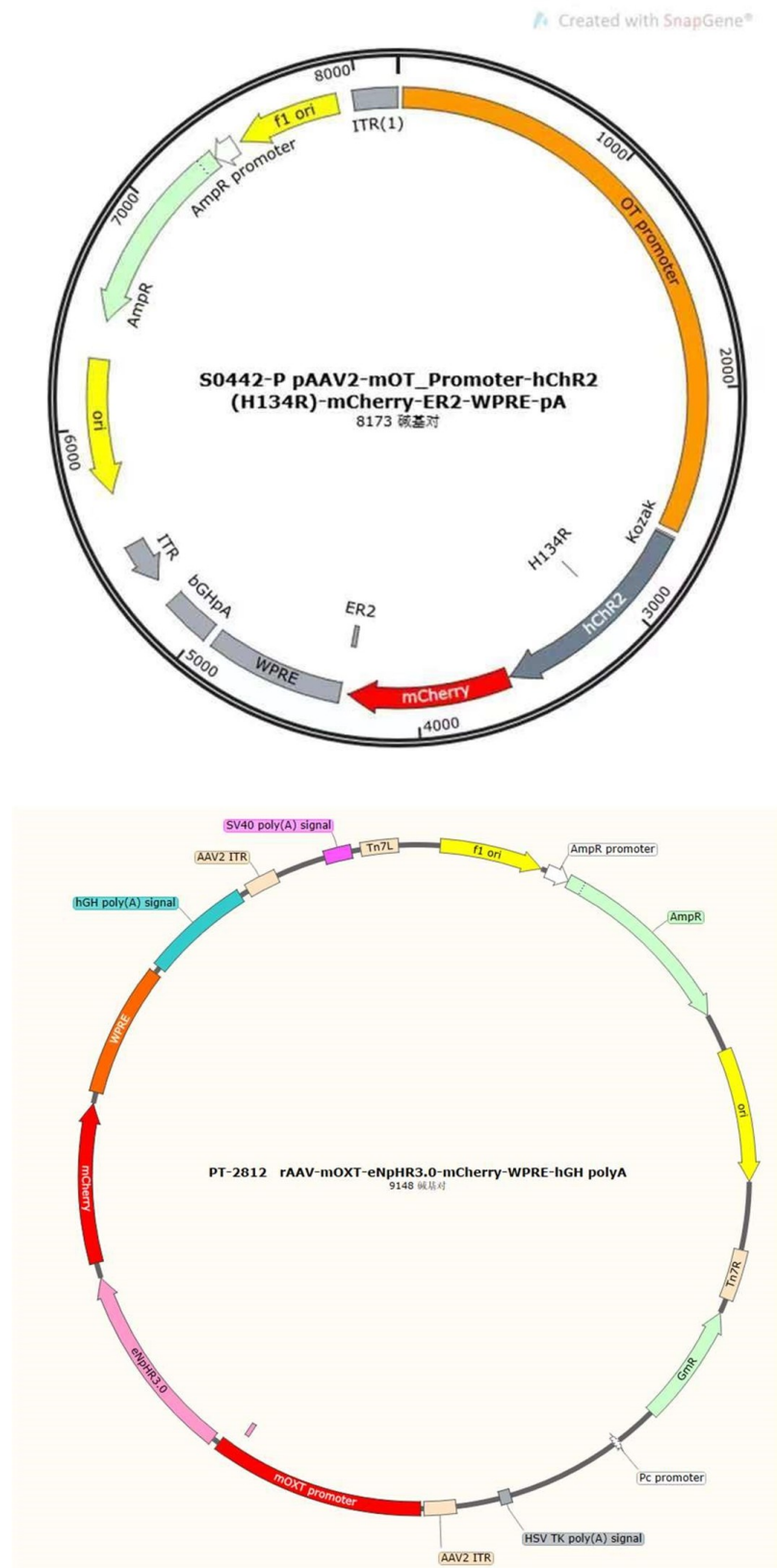
As our response to reviewer 2, animals generally exhibit three types of behavioral responses toward pups, and data on the percentage of these different behavioral types occurring in the group will be included in another study in our lab. The reviewer's suggestion of scoring the behaviors is an inspiring idea that will help us to more fully parse these behaviors. Mandarin voles were captured from the wild in Henan, China. The experimental subjects were F2 generation voles reared in the Experimental Animal Centre of Shaanxi Normal University. In our observations, pup care and infanticide behaviors were conserved across several pup exposures, especially pup care behaviors, whereas for infanticide behaviors we did not conduct more pup exposures in order to protect the pups.

Optogenetics with the oxytocin promoter virus is a nice advance here. More details about their preparation and methods should be in the main text, and not simply relegated to the methods section. For optogenetic stimulation in Figure 2, how were the stimulation parameters chosen? There is a worry that oxytocin neurons can co-release other factors—are the authors sure that oxytocin is being released by optogenetic stimulation as opposed to other transmitters or peptides, and acting through the oxytocin receptor (as opposed to a vasopressin receptor)?

As reviewer suggested, more detailed information about virus construction and choice of optogenetic stimulation parameter have been added in the revised manuscript. The details about the construction of CHR2 and mCherry viruses used in optogenetic manipulation can refer to a previous study in which they constructed an rAAV-expressing Venus from a 2.6 kb region upstream of OT exon 1, which is conserved in mammalian species (Knobloch et al., 2012). For details about construction of the eNpHR 3.0 virus, expression of the vector is driven by the mouse OXT promoter, a 1kb promoter upstream of exon 1 of the OXT gene, which has been shown to induce cell type-specific expression in OXT cells (Peñagarikano et al., 2015). Details about the construction of OT1.0 sensor can be referred to the research of Professor Li's group (Qian et al., 2023). The mapping of the viral vectors and OT1.0 sensor is shown below.

The optogenetic stimulation parameters were used based on a previous study (He et al., 2021). However, our description of the parameters in the experiment is still not in detail, so some information about optogenetic stimulation parameters has been added in the method. In pupdirected pup care behavioral test, light stimulation lasted for 11 min. Parameters used in optogenetic manipulation of PVN OT neurons were ~ 3 mW, 20 Hz, 20 ms, 8 s ON and 2 s OFF and parameters used in optogenetic manipulation of PVN OT neurons projecting to mPFC were ~ 10 mW, 20 Hz, 20 ms, 8 s ON and 2 s OFF to cover the entire interaction. We performed fiber photometric experiments to determine the role that OT plays in behavior, and these results were able to support each other with optogenetic experiments. In addition, we further confirmed the role of optogenetic manipulation on OT release in combination with optogenetic inhibition and OT1.0 sensors (Supplementary Data Fig. 2). It has been previously shown that OT is able to act specifically on OTR in mPFC-PL (Sabihi et al., 2017). Our study focuses on oxytocin neurons as well as oxytocin release, and more research is needed to construct a more complex and complete network regarding the involvement of the OTR and other factors in the mPFC in these behaviors.

Author response image 1.



Author response image 2.



Given that they are studying changes in latency to approach/attack, having some controls for motion when oxytocin neurons are activated or suppressed might be nice. Oxytocin is reported to be an anxiolytic and a sedative at high levels.

As our response to reviewer 1, to exclude effects of locomotor activity on pup directed behaviors, we also investigated effect of optogenetic manipulations on the locomotor activity of experimental animals and found that optogenetic manipulation did not change levels of locomotor activity (Supplementary Data Fig. 6).

The OT1.0 sensor is also amazing, these data are quite remarkable. However, photometry is known to be susceptible to motion artifacts and I didn't see much in the methods about controls or correction for this. It's also surprising to see such dramatic, sudden, and large-scale suppression of oxytocin signaling in the mPFC in the infanticidal animals - does this mean there is a substantial tonic level of oxytocin release in the cortex under baseline conditions?

The optical fiber recording system used in the present study can automatically exclude effects of motion artifacts by simultaneously recording signals stimulated by a 405nm light source. As shown in the formula below, the z-score data were calculated and presented, and the increase and decline of the OT signal is a trend relative to the baseline. For a smooth baseline, the decreasing signal is generally amplified after calculation. In our experiments combining optogenetic inhibition and OT1.0 sensors, we were able to find that there was a certain level of OT release at baseline, on which there was room for a decrease in the signal recorded by the OT1.0 sensor.

$$F_0 = \overline{V_{basal}} \quad \frac{\Delta F}{F_0} = \frac{V_{signal} - F_0}{F_0 - V_{offset}} \quad \sigma_F = \text{STD}(V_{basal}) \quad \text{Z-score} = \frac{V_{signal} - F_0}{\sigma_F}$$

Figure 5 is difficult to parse as-is, and relates to an important consideration for this study: how extensive is the oxytocin neuron projection from PVN to mPFC?

AAV11-Ef1a-EGFP virus can infect fiber and retrogradely reach to cell body, thus this virus can be used to retrogradely trace neurons. We injected the this virus (green, AAV11-Ef1aEGFP) in the mPFC and observed virus infected and OT (red) positive neuron in the PVN (Yellow), and we also counted the OT neurons that project from PVN to mPFC and found that approximately 45.16% and 40.79% of cells projecting from PVN to the mPFC were OT-positive, and approximately 18.48% and 18.89% of OT cells in the PVN projected to the mPFC in females and males, respectively (Supplementary Data Fig. 4).

In Figures 6 and 7, the authors use the phrase 'projection terminals'; however, to my knowledge, there have not been terminals (i.e., presynaptic formations opposed to a target postsynaptic site) observed in oxytocin neuron projections into target central regions.

According your suggestion, we replaced the 'terminals' with 'fibers' to describe it more accurately..

Projection-based inhibition as in Figure 7 remains a controversial issue, as it is unclear if the opsin activation can be fast enough to reduce the fast axonal/terminal action potential. Do the authors have confirmation that this works, perhaps with the oxytocin GRAB OT sensor?

Thanks for your suggestion. We measured the OT release using OT1.0 sensors when the OT neuron projections in the mPFC were optogenetically inhibited. The result showed that optogenetic inhibition of OT neuron fibers in the mPFC significantly reduced OT release that validate the method of projection-based inhibition (Supplementary Data Fig. 5).

As females and males had similar GRAB OT1.0 responses in mPFC, why would the behavioral effects of increasing activity be different between the sexes?

In the present study, females released higher levels of OT into the mPFC (Figure 4 d, e) than males upon occurrence of different behaviors. In addition, females already exhibited more rapid approach and retrieval of pups than male before the optogenetic activation this may be the reason no effects of this manipulation were found in female.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) Check for spelling and grammar errors throughout.

Thanks to the reviewer's suggestion, we have checked and revised the article.

(2) Report effect sizes for all significant findings to allow evaluation of practical significance.

As reviewer suggested, we have added effect size both in the source data and in the main text, including d, η^2 and odds ratio.

(3) Provide detailed information on the imaging parameters and analysis methods used in the Fos study.

The imaging parameters and analysis details for the Fos study have also been added in the revised manuscript. Brain slices of 40 μm thick were collected consecutively on 4 slides, each slide had 6 brain slices spaced 160 μm apart from each other. PVN area were determined

based on the Allen Mouse Brain Atlas and our previous study, and Fos, OT and merged positive neurons were counted.

(4) Compare the Fos colocalization results with previous studies examining PVN Fos and maternal/paternal care to contextualize the findings.

Our result about Fos and OT colocalization is consistent with previous study. In a previous study on virgin male prairie voles, OT and Fos colabeled neurons in the PVN increased after exposure to conspecific pups and experiencing paternal care (Kenkel et al., 2012). In another study of prairie voles, OT and c-fos colabeled neurons in PVN significantly increased after becoming parents which may be due to a shift from virgin to parents (Kelly et al., 2017).

(5) Discuss the limitations of the study, such as the potential effects of manipulating OT neurons on the release of other transmitters or the influence of other neurochemicals or brain regions on pupdirected behaviors, especially in females.

We agree with reviewer's suggestion that several limitations should be discussed. Although we used a virus strategy to specifically activate or inhibit PVN OT neurons, other neurochemical may also be released during optogenetic manipulations because OT neurons may also release other neurochemicals. In one of our previous studies, activation of the OT neuron projections from the PVN to the VTA as well as to the Nac brain also altered pup-directed behaviors, which may also be accompanied by dopamine release (He et al., 2021). In addition, backpropagation of action potentials during optogenetic manipulations may also causes the same behavioral effect as direct stimulation of PVN OT cells. These effects on pup-directed behaviors should also be investigated further in the future study.

(6) Address the possibility of back-propagation of action potentials in the optogenetic manipulations causing the same behavioral effects as PVN OT cell stimulation.

We agree with the reviewer's opinion hat optogenetic manipulation may possibly induce back-propagation of action potentials that may result in same behavioral effects as OT cell stimulation. We will pay attention to this issue in the future study.

(7) Investigate whether changes in locomotor behavior could explain the observed effects on pupdirected behaviors.

To exclude effects of locomotor activity on pup directed behaviors, we also investigated effect of optogenetic manipulations on the locomotor activity of experimental animals and found that optogenetic manipulation did not change levels of locomotor activity (Supplementary Data Fig. 6).

(8) Report the percentage of PVN->mPFC neurons labeled that were OT-positive.

AAV11-Ef1a-EGFP virus can infect fiber and retrogradely reach to cell body, thus this virus can be used to retrogradely trace neurons. We injected this virus (green, AAV11-Ef1a-EGFP) in the mPFC and observed virus infected and OT (red) positive neuron in the PVN (Yellow), and we also counted the OT neurons that project from PVN to mPFC and found that approximately 45.16% and 40.79% of cells projecting from PVN to the mPFC were OT-positive, and approximately 18.48% and 18.89% of OT cells in the PVN projected to the mPFC in females and males, respectively (Supplementary Data Fig. 4).

(9) Directly compare the sexes in the behavioral analysis and discuss any potential sex differences.

We agree with the reviewer's suggestion and have added comparisons between two sexes and discussion about relevant results.

(10) If available, report and discuss the OT expression levels and the number of OT neurons manipulated in each sex.

In the present study, we have counted the number of OT cells, but did not measure the level of OT expression using WB or qPCR. In addition, the percentages of CHR2(H134R) and eNpHR3.0 virus infected neurons in total OT positive neurons were presented (Supplementary Data Fig. 7), but we did not know how many cells were actually manipulated during the optogenetic experiment.

(11) Expand the discussion to include what could be regulating or interacting with the OT circuit under investigation, particularly in females where the effects were less pronounced.

As the reviewers suggested, we have also added relevant discussion. In addition to OT neurons, OTR neurons may also regulate behavioral responses to pups. In a study of virgin female mice pup exposure was found to activate oxytocin and oxytocin receptor expressing neurons (Okabe et al., 2017). Other brain regions such as preoptic area (POA) may also be involved in parental behaviors. For example, virgin female mice repeatedly exposed to pups showed shorter retrieval latencies and greater c-Fos expression in the preoptic area (POA), concentrations of OT in the POA were also significantly increased, and the facilitation of alloparental behavior by repeated exposure to pups occurred through the organization of the OT system (Okabe et al., 2017). A recent study suggests that OT of the PVN is involved in the care of pups by male voles (He et al., 2021). This study suggests that PVN to ventral tegmental area (VTA) OT projections as well as VTA to nucleus accumbens (NAc) DA projections are involved in the care of pups by male voles. Inhibition of OT projections from the PVN to the VTA reduces DA release in the NAc during licking and grooming of pups (He et al., 2021).

Reviewer #2 (Recommendations For The Authors):

A few additional things the authors may want to consider:

(1) I don't understand the subject numbers in the peripheral OT study data shown in Figure 8. Panels p and q have 69 females shown and 50 males. Was there a second, much larger, IP injection study conducted that was different than the subjects shown in panels a-o that had ~5 subjects per treatment group per sex?

Sorry for the confusing. More animals were used to test effects of OT on infanticide behaviors in our pre-test. These data combined with data from formal pharmacological experiment were presented in Fig. 8p, q. After OT treatment, the changes in detailed and specific behaviors were only collected in several animals. We have clarified that in the revised manuscript.

(2) The authors suggest higher baseline OT release in the female mPFC, which makes sense and helps explain some of their results. It seems that the data in Figure 1 show what is probably no sex difference in OT cell numbers in the PVN of Mandarin voles, which is unlike the old studies in mice or rats. If readers look at the data in Figure 1 showing what seems to be no sex difference in OT cell number, the authors' argument in the discussion about mPFC OT release levels higher in females would be inconsistent with their own data shown. The authors have the brain sections they need to help support or undermine this argument in the discussion, so maybe it would be useful to analyze the

OT cell numbers across the PVN and report it in this paper or briefly mention it in the discussion.

We compared the numbers of OT neurons, activated OT neurons (OT and Fos doublelabeled neurons) and level of OT release between males and females. We found that females have more activated OT neurons (Figure1, d, g) and released higher levels of OT into the mPFC (Figure 4 d, e) than males. This part has been added in the result and discussion. The inconsistency of the OT cell numbers with previous studies may be due to the method of cell counting, as we did not count all slides consecutively.

(3) The discussion suggests visual cues are involved in mPFC OT release relevant for pup care or infanticide, but this is a very odd claim for nocturnal animals that live and nest with their pups in underground burrows.

Sorry for the confusing. Here, we cited the finding in mice that activation of PVN OT neurons induced by visual stimulation promoted pup care to support our finding that the activity of OT cells of the PVN is involved in pup care, rather than to illustrate the role of visual stimulation in voles. We have clarified that in the revised manuscript.

(4) The lack of decrease in mPFC OT release in the 2nd and 3rd approaches to pups is probably because the release was so high after the 1st approach that it didn't have time to drop before the subsequent approaches. The authors don't state how long those between-approach intervals were on average to help readers interpret this result.

As described in our methods, we spaced about 60 s between each behavioral test to allow the signal return back to the baseline level.

(5) Do PVN-mPFC OT somata collateralize to other brain sites? Could mPFC terminal stimulation activate entire PVN cells and every site they project to? A caveat could be mentioned in the discussion if there's support for this from other optogenetic and PVN OT cell projection studies.

We verified the OT projections from PVN to mPFC, to validate the optogenetic manipulation of this pathway, but did not investigate whether the OT neurons projecting from PVN to mPFC also project collaterally to other brain regions. It is suggested that mPFC terminal stimulation only activate PVN OT cells projecting mPFC, whether other OT neurons were activated remains unclear.

(6) I don't see an ethics statement related to the experiments obviously having to involve pup injury or death. Nothing is said in methods about what happened after adult subjects attacked pups. I assumed the tests were quickly terminated and pups euthanized.

In case the pups were attacked, we removed them immediately to avoid unnecessary injuries, and injured pups were euthanized.

(7) The authors could be more specific about what psychological diseases they refer to in the abstract and elsewhere that are relevant to this study. Depression? Rare cases of psychosis? Even within the already rare parental psychosis, infanticide is tragic but rare.

Infanticide is caused by a variety of factors, mental illness, especially depression and psychosis, is often a very high risk factor among them (Milia & Noonan, 2022; Naviaux, Janne, & Gourdin, 2020). In human, infanticide has been used to refer to the killing, neglect or abuse of newborn babies and older children (Jackson, 2006). Here, we believe that research on the

neural mechanisms of infanticide can also contribute to the understanding and treatment of attacks on children, physical and verbal abuse, and direct killing of babies.

(8) Figure 8 - in one case the "*" is a chi-square result, correct?

Thanks for your careful checking. In Figure 8p, q, we applied the chi-square test and added it in the legend.

Reviewer #3 (Recommendations For The Authors):

The only other thing is a typo on line 135: the authors mean 'stimulation' instead of 'simulation'.

Corrected.

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